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REVIEW

A Systematic Review on Deep Learning for Atmospheric Correction of Satellite Images

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Abstract

Deep learning (DL) has recently emerged as a transformative approach for atmospheric correction (AC) in satellite remote sensing, addressing the limitations of traditional image-based and physics-based models. This paper presents a comprehensive and structured systematic review of DL-based AC methods, aiming to bridge the existing knowledge gap by consolidating insights across 30 peer-reviewed studies. The reviewed methods are categorised into physics-aware and physics-agnostic frameworks based on whether they require auxiliary information such as atmospheric and geometric parameters and top of atmospheric reflectance. In addition, the paper compiles commonly used datasets, including satellite, simulated, and field-measured data and evaluation metrics. It also identifies critical DL model assumptions, such as fixed atmospheric profiles, clear-sky conditions, and Lambertian surface approximations. The paper identifies key challenges and proposes future directions by synthesising current research. To our knowledge, this is the first systematic review of DL-based AC. This work is a foundational resource for researchers and practitioners, guiding the development of robust, interpretable, and scalable DL-based AC models.

1 Introduction

Remote sensing has emerged as a vital tool for Earth observation, supporting diverse applications such as environmental monitoring, disaster management, land-use classification [1], and climate studies. Satellite-based remote sensing enables systematic, high-resolution data acquisition over large geographic areas, providing critical insights into surface and atmospheric processes. However, the raw satellite signals captured at the Top of the Atmosphere (TOA) are significantly affected by atmospheric interference, including scattering and absorption by gases, aerosols, and water vapour, as shown in Fig. 1. A satellite at the top of the atmosphere detects both the reflected and scattered radiation, highlighting the key steps in remote sensing measurements.

These interactions distort the spectral signatures of surface features called Bottom of Atmosphere (BOA) or surface reflectance (SR). Atmospheric correction (AC) is one of the most crucial preprocessing steps, as it transforms TOA radiance into BOA or SR, allowing for accurate surface property retrievals. For example, in Fig. 2, image (b) illustrates the result of AC applied to image (a), demonstrating the transformation from TOA radiance to BOA. The corrected image exhibits enhanced contrast and more realistic surface colours, facilitating an improved interpretation of land and water features.

Several other preprocessing steps are required before remote sensing imagery can be used for quantitative applications. These include geometric correction to align the imagery spatially, radiometric calibration to convert raw digital numbers into physically meaningful radiance or reflectance values, and cloud detection and masking to remove cloud-contaminated pixels. Among these, efforts to improve satellite data's scientific usability and interoperability have led to developing Analysis Ready Data (ARD) products, which standardise these preprocessing steps across missions and sensors. ARD aims to deliver spatially aligned datasets, radiometrically calibrated and atmospherically corrected,

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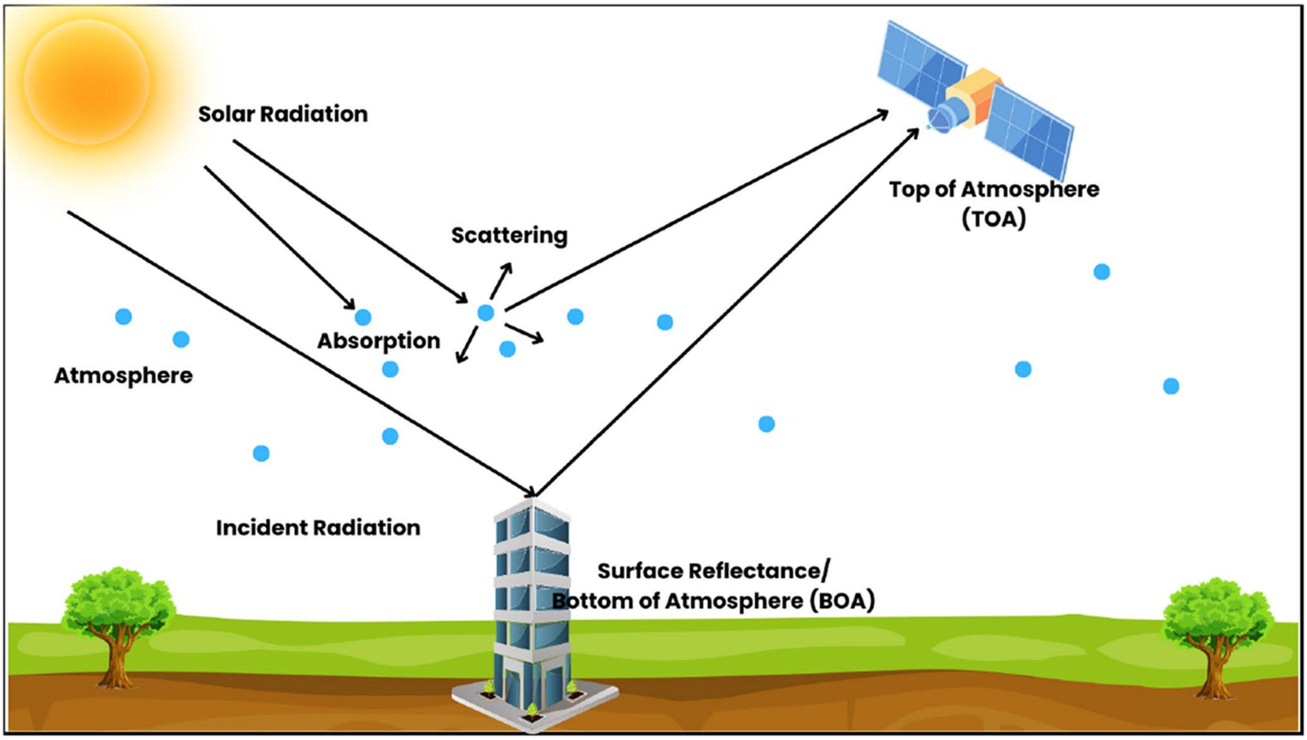
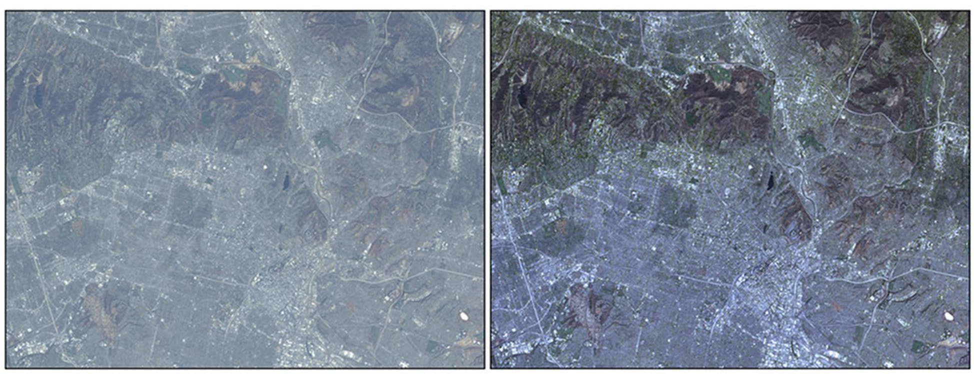


Fig. 1 Distortions impacting the TOA signal received by the satellite

Fig. 2 Left: landsat 8 TOA reflectance image, right: landsat 8 BOA reflectance image for an area over Los Angeles, California, path 41 row 36 [2]



with quality flags and metadata for traceability. Global initiatives such as the Committee on Earth Observation Satellites (CEOS) have spearheaded the standardisation of ARD specifications [3–5]. For example, the CEOS-ARD guidelines define metadata conventions requirements, uncertainty quantification, quality assurance flags, and processing traceability to promote transparency and reproducibility [6]. Major remote sensing platforms such as Landsat Collection 2 SR Tier 1, Sentinel-2 Level-2A, and USGS ARD products now conform to these specifications [7].

As the demand for high-resolution, temporally consistent, and application-ready remote sensing data grows, robust AC has become indispensable [8] as it aims to compensate for atmospheric effects, transforming TOA radiance into SR, which more accurately represents the Earth's surface properties. Accurate SR retrieval is fundamental to numerous remote sensing applications, including vegetation monitoring [9, 10], crop type classification [11], water quality assessment [12, 13], climate analysis [14, 15], and land cover change detection [16, 17].

AC is inherently challenging due to the dynamic nature of atmospheric conditions. The variability in aerosol concentrations, water vapour content, and gas composition introduces the correction process's uncertainties. Many correction algorithms rely on atmospheric parameters such as Aerosol Optical Depth (AOD) and Column Water Vapour (CWV) content, which are often unavailable or difficult to measure with high accuracy. Additionally, the spectral response functions of different satellite sensors vary, necessitating sensor-specific calibration models, which restricts cross-sensor applicability. Moreover, the complex interaction between the atmosphere and heterogeneous surface types, especially in urban or mixed land cover regions, adds further non-linearity to the problem.

AC has traditionally relied on image-based and physics-based methods to retrieve accurate SR from remotely sensed data. With the advent of Deep Learning (DL), data-driven approaches for AC have emerged as powerful alternatives. The following subsections provide a detailed overview of three paradigms: image-based, physics-based, and DL-based AC techniques.

1.1 Image-Based Atmospheric Correction

Image-based methods rely on statistical relationships derived from observed radiance values and corresponding SR measurements. These methods utilise reference targets or in-scene characteristics to establish empirical models that relate observed radiance to SR. Dark Object Subtraction (DOS) [18] estimates atmospheric contributions by assuming specific spectral properties of the scene, including dark objects with nearly zero reflectance. Empirical Line Calibration (ELC) [19] and Internal Average Reflectance (IAR) use scene statistics to approximate atmospheric effects. Histogram matching is another widely used image-based approach that attempts to standardize the radiometric properties of images based on reference histograms. While these methods are relatively simple to implement and computationally efficient, their reliance on suitable reference targets or invariant features, such as dark objects, bright surfaces, or spectrally stable regions, makes them less robust in complex atmospheric conditions. In heterogeneous landscapes or under variable atmospheric scenarios (e.g., varying aerosol loads, cloud contamination, or high humidity), identifying consistent reference features becomes difficult, leading to increased uncertainty in AC. Consequently, the accuracy and generalisability of these methods are often compromised, particularly across different seasons, sensor types, and geographic regions.

1.2 Physics-Based Atmospheric Correction

Physics-based AC methods rely on radiative transfer principles to model and remove atmospheric effects from satellite-observed signals, thereby retrieving accurate SR. The Radiative Transfer Equation (RTE) lies at the core of these methods, which mathematically describes how electromagnetic radiation is scattered, absorbed, and transmitted as it propagates through the atmosphere.

However, solving the RTE numerically for every pixel in a high-resolution satellite image is computationally prohibitive. Instead, radiative transfer models like MODTRAN simulate the RTE using pre-defined atmospheric profiles. Such models take as input a set of atmospheric parameters, such as AOD, CWV, ozone concentration, and geometric parameters like solar zenith angle, sensor viewing angle, and relative azimuth angle. These parameters define the state of the atmosphere and viewing geometry at the time of image acquisition. Based on these inputs, these models calculate key radiative transfer components: path radiance (radiation scattered into the sensor from the atmosphere), direct and diffuse transmittance (representing how much radiation travels unimpeded to or from the surface), and spherical albedo (accounting for light reflected within the atmosphere).

These quantities are then used in a simplified version of the RTE that relates the observed TOA radiance to SR as given in equation 1 [20].

$$L(\lambda) = L_{\mathrm{pa}}(\lambda) + T(\lambda) \cdot \frac{\rho(\lambda) \cdot E_{\mathrm{s}}(\lambda) \cdot \cos(\theta_0)}{\pi} \tag{1}$$

$L(\lambda)$ Top-of-Atmosphere radiance at wavelength λ observed by the satellite sensor.
$L_{\mathrm{pa}}(\lambda)$ Path radiance due to atmospheric scattering along the sensor's line of sight.
$T(\lambda)$ Total atmospheric transmittance (product of upward and downward transmittance) at wavelength λ.
$\rho(\lambda)$ SR/BOA reflectance to be retrieved.
$E_{\mathrm{s}}(\lambda)$ Solar irradiance at the TOA at wavelength λ.
θ_0 Solar zenith angle, the angle between the sun's rays and the vertical at the surface location.
π Normalization constant in the Lambertian surface assumption.

To improve computational efficiency, operational AC algorithms precompute solutions to the RTE across a comprehensive range of parameter combinations and store the outputs in multidimensional Look-Up Tables (LUTs). These LUTs serve as efficient surrogates for full radiative transfer simulations, enabling the rapid estimation of SR through interpolation techniques. The construction of LUTs involves running the RTE for different configurations of atmospheric and surface parameters, generating an extensive database of TOA radiance or reflectance values. During operational correction, the satellite-observed TOA radiance is matched against the LUTs to retrieve the most probable SR. This LUT-based approach accelerates the inversion process, making it feasible for large datasets. Fig. 3(a) illustrates a block diagram of the physics-based AC approach.

Several well-established radiative transfer models serve as the computational backbone for these physics-based approaches. The 6S (Second Simulation of the Satellite Signal in the Solar Spectrum) model [20] provides an accurate

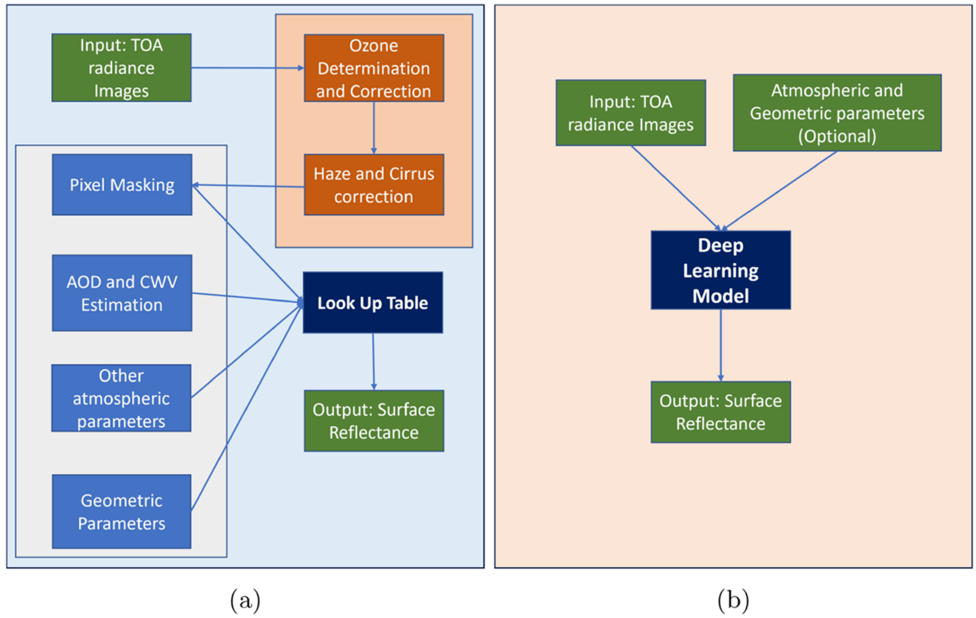


Fig. 3 (**a**): physics-based model and (**b**): DL-based model for atmosperic correction

simulation of TOA radiance under various atmospheric conditions and sensor geometries. MODTRAN (MODerate Resolution Atmospheric TRANsmission) [21] offers highly detailed spectral simulations of atmospheric transmission, path radiance, and absorption across a wide wavelength range. Atmospheric and Topographic Correction (ATCOR) [22] further extends the modelling framework by incorporating terrain effects, allowing for more accurate correction in areas with complex topography.

Despite their advantages, physics-based methods present several challenges. The requirement for detailed and accurate atmospheric parameters presents a significant challenge, particularly in regions or periods lacking concurrent in situ or auxiliary measurements. Errors or uncertainties in input parameters such as AOD or CWV can propagate through the correction process, leading to inaccurate SR estimates. Moreover, the high computational demand associated with LUT generation and interpolation can hinder scalability, particularly for large-scale or near-real-time processing. Additionally, building physics-based models often demands a strong understanding of radiative transfer principles and atmospheric physics, which can limit accessibility for operational or non-specialist users.

1.3 Deep Learning Based Atmospheric Correction

The limitations of traditional physics-based methods, including their dependency on auxiliary data, computational inefficiency, and challenges in addressing diverse atmospheric conditions, have driven the exploration of DL models for AC. DL presents an opportunity to achieve accuracy comparable to physics-based methods while improving computational efficiency and reducing reliance on externally measured atmospheric parameters. Unlike physics-based methods, which require explicit modelling of atmospheric interactions, DL models learn complex spectral and spatial relationships directly from data, enabling them to generalise across varying atmospheric conditions without the need for detailed atmospheric input. When trained on large, diverse datasets, DL models exhibit strong generalisation capabilities across geographic regions [23, 24] and atmospheric scenarios [25, 26]. Their integration of learning and inference into a unified framework enables continuous refinement, making them more resilient to biases introduced during separate model training and prediction stages.

Additionally, the accessibility of DL approaches makes them attractive for broader adoption. Unlike physics-based methods that require domain expertise in radiative transfer modelling, non-experts can employ DL models, provided sufficient training data is available. Furthermore, DL approaches can be optimised for real-time processing, making them well-suited for large-scale remote sensing applications. Once trained, these models can deliver near real-time inference and be deployed on edge devices, enabling responsive applications in disaster management [27], agricultural assessment [28], and environmental monitoring [29]. Advances in hardware acceleration, including GPUs and TPUs, have further enhanced the scalability of DL models, allowing efficient processing of high-resolution satellite imagery without the bottlenecks associated with radiative transfer models.

Fig. 3(b) presents the conceptual framework of a DL-based AC approach. DL models take TOA radiance as input in this framework and optionally incorporate ancillary geometric and atmospheric parameters to enhance learning. Unlike physics-based methods, which explicitly simulate atmospheric interactions, DL models learn the complex, non-linear mapping from TOA radiance to SR directly from data. By training on large datasets containing paired TOA and BOA observations, these models can generalise across diverse atmospheric and geographic conditions.

The qualitative effectiveness of such an approach is demonstrated in Fig. 4, which showcases the performance of the Season-Aware AC Network (SAAC-Net) [30]. In this example, the BOA reflectance images predicted by SAAC-Net for Landsat 8 imagery closely resemble those produced by the physics-based Land Surface Reflectance Code (LaSRC) [31] model, highlighting the potential of DL models to approximate traditional radiative transfer-based corrections.

Despite these advancements, research in DL-based AC remains fragmented. There is a lack of comprehensive reviews that consolidate existing methodologies, evaluate progress, and identify open challenges. This gap hinders a holistic understanding of the field's current and future trajectory. This systematic review addresses that gap by analysing DL approaches for the AC of satellite imagery. It discusses commonly used datasets, training strategies, and evaluation metrics while highlighting best practices and recurring challenges such as data dependency, generalisation, and model transferability. The review also explores emerging directions, including hybrid physics-informed DL models, domain adaptation techniques, and the creation of standardised benchmark datasets to support model comparison and reproducibility. By addressing both methodological and operational aspects, this review aims to guide future research and development efforts toward more accurate, efficient, and scalable AC solutions aligned with the growing demands of modern Earth observation systems. To our knowledge, this is the first comprehensive review focused exclusively on DL-based AC approaches.

The rest of the paper is organised as follows. Sect. 2 explains the methodology used for identifying, screening, and selecting the literature. Sect. 3 discusses the different types of datasets such as satellite, simulated or field data used in DL-based AC. This section also explains standard evaluation metrics. Sect. 4 covers training strategies, approaches, design principles and assumptions for DL based AC models. The open issues and future research directions are covered in Sect. 5, with Sect. 6 concluding the paper.

2 Materials and Methods

This study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [32] 2020 statement protocol to ensure a structured and reproducible methodology for identifying, screening, and selecting relevant literature. A comprehensive literature search was conducted across multiple scientific databases, including Scopus[1], Google Scholar[2], and IEEE Xplore[3]. These databases were selected due to their broad coverage of

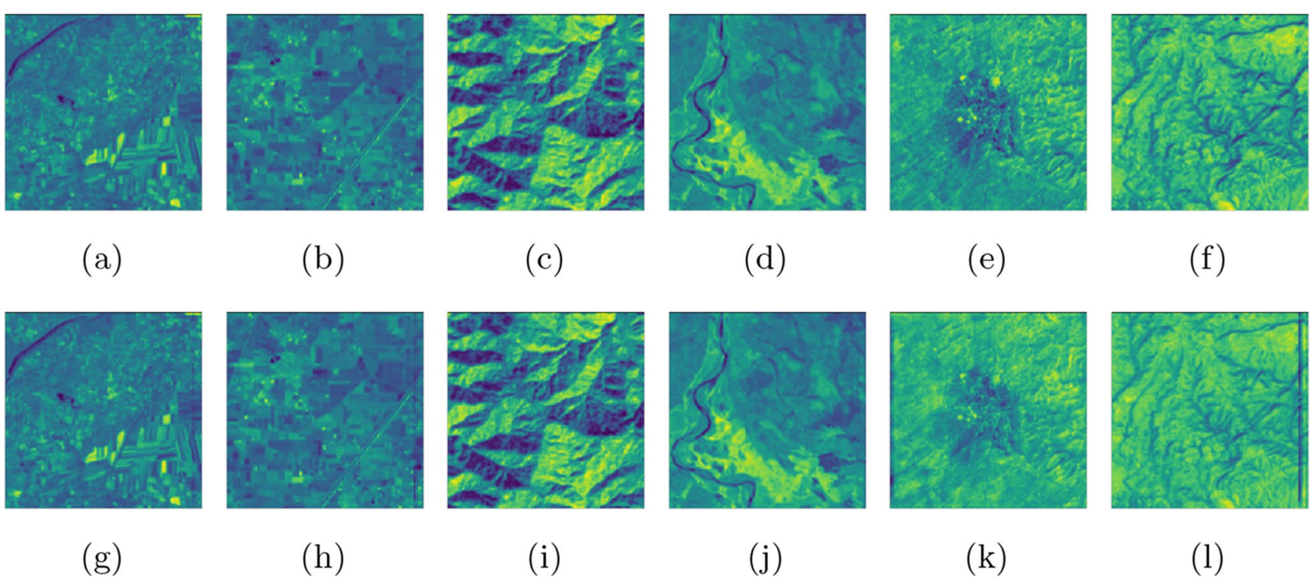


Fig. 4 The first row shows the landsat 8 BOA images (**a-f**), and the second row shows SAAC-Net-generated images (**g-l**) for (following left to right) urban land, cropland, deciduous forest, Evergreen forest, fallow land, and wasteland respectively [30]

[1] https://www.scopus.com/

[2] https://www.scholar.google.com/

[3] https://www.ieeexplore.ieee.org/

high-quality peer-reviewed research in remote sensing, DL, and AC. The search was conducted up to April of 2025, employing tailored keyword strings to optimise the relevance and consistency of the retrieved studies. The search strategy involved querying within the article title, abstract, and keywords using specific keyword combinations to target studies on DL-based AC. The following keyword combinations were used:

- "Deep learning" OR "Machine learning" OR "Neural networks"
- "Atmospheric Correction" OR "Surface Reflectance"
- "Remote sensing"
- "Radiative transfer model" OR "Radiative transfer code"

The search was limited to peer-reviewed journal articles and conference proceedings to ensure the inclusion of relevant and recent advancements in the field. Using the above keywords, we searched Scopus, IEEE, and Google Scholar and retrieved 146, 124, and 62 articles. Then, 82 duplicate records were removed, which left 250 articles for screening. We manually screened these articles and removed articles based on the following Exclusion Criteria (EC).

- EC1: Publications unrelated to atmospheric correction/surface reflectance retrieval/radiative transfer model.
- EC2: Publications do not use Machine Learning (ML) or DL.
- EC3: Publications use the terms atmospheric correction/surface reflectance retrieval/radiative transfer model, and ML/DL, but ML/DL has not been used.
- EC4: Based on the journal impact factor and the citations the publication received.
- EC5: Publications published before 2015.
- EC6: Publications not in English.

Since our paper focuses on AC or SR retrieval of satellite images using ML or DL, we excluded publications irrelevant to AC, SR retrieval, or radiative transfer models (EC1). Additionally, studies that focused on AC, SR retrieval, or radiative transfer models but did not employ ML or DL (EC2) were excluded. After applying EC1 and EC2, 118 publications remained. Among these, some studies involved AC, SR retrieval, or radiative transfer models and mentioned ML/DL but did not use these techniques for AC or SR retrieval. These were excluded as well (EC3). Furthermore, publications in journals that had an impact factor of less than 1 and having no citations (EC4), were published before 2015 (EC5), or were not in English (EC6), were also removed from consideration. The PRISMA flow diagram in Fig. 5 illustrates the identification, screening, and inclusion-exclusion process with many articles.

Figure 6 illustrates the publication trend over time, showing the number of research papers published per year within the domain of study. The bar chart represents the annual publications count, while the overlaid line plot highlights the overall trend. A steady increase in research activity is observed from 2017 onward, with a notable surge in publications peaking in 2023. This suggests that the field is new, and the interest in this field has been increasing in the past few years. This trend indicates that the domain has gained momentum in recent years, reflecting increasing recognition of its importance and potential for further exploration.

For each selected study, relevant details were extracted and categorised to facilitate a structured comparison of DL-based AC methodologies. The extracted information is categorised as follows:

1. Auxiliary Data: Specifies the type of input used for the DL model, including atmospheric parameters and geometric parameters in addition to TOA radiance data.
2. Physics Model: The underlying physics-based model used by the satellite or a physics-based model used to generate synthetic data.
3. Dataset used: Indicates whether the study used satellite imagery or simulated data generated from radiative transfer models such as MODTRAN [21], Accurate Radiative Transfer (ACCuRT) [33], or Soil Canopy Observation of Photosynthesis and Energy fluxes (SCOPE) [34].
4. Multispectral/Hyperspectral: Indicates whether the dataset consists of multispectral or hyperspectral data
5. Spatial Processing: Differentiates between pixel-based spatial processing, which processes each pixel independently, and image-based processing, which takes the whole TOA image as input and considers spatial dependencies.
6. Target Environment: Defines the scope of the study, highlighting its relevance to specific target environments such as land, coastal zone, inner water, and open ocean for which the model has been designed.
7. Model used: Identifies the specific DL architecture applied, such as convolutional neural networks (CNNs), Neural networks (NNs), or encoder-decoder architectures.
8. End-to-End model: Examines whether the model performs a complete AC process without requiring additional post-processing steps. If the model's output generates the SR values, it has been classified as an End-to-End model; otherwise, it is not.
9. Results: Provides key findings from the study, summarising the model's accuracy and overall performance.

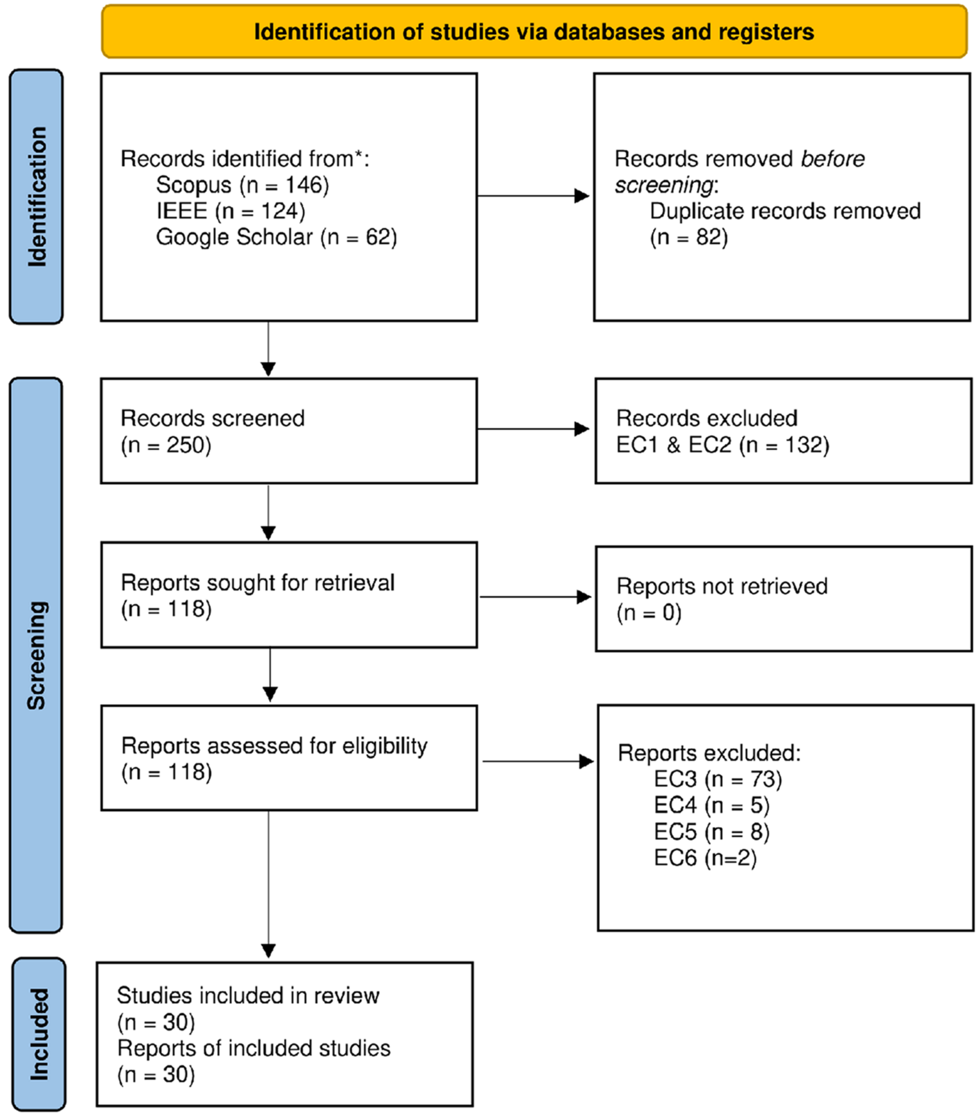


Fig. 5 PRISMA flow diagram for selection of studies considered in the review

10. Validation using Ground Data: Indicates whether ground-truth measurements were used for validation and specifies the type of field data collected, if applicable.

3 Dataset and Evaluation Metrics

The development and validation of DL-based AC models critically depend on the quality and diversity of datasets. These datasets generally fall into three broad categories: satellite datasets, simulated datasets, and field measurements. Satellite multispectral and hyperspectral datasets provide real-world observations across different geographic and atmospheric conditions. Simulated datasets, generated using radiative transfer models, offer controllable data for model training under fixed atmospheric scenarios. Field measurements are the reference standard for validating AC models and include data collected through global ground-based networks and targeted field campaigns. This section describes these datasets and outlines the evaluation metrics commonly used to assess model performance across studies, such as Root Mean Square Error (RMSE), correlation coefficient, and Mean Absolute Percentage Error (MAPE).

3.1 Satellite Datasets

The effectiveness of DL-based AC models largely depends on the selection and quality of training datasets. These datasets can be categorised into multispectral and hyperspectral sources, each with distinct characteristics influencing model performance. Multispectral datasets, such as Sentinel-2 [35], MODIS [36], and Landsat [37], are widely used due to their extensive spatial and temporal coverage. Sentinel-2,

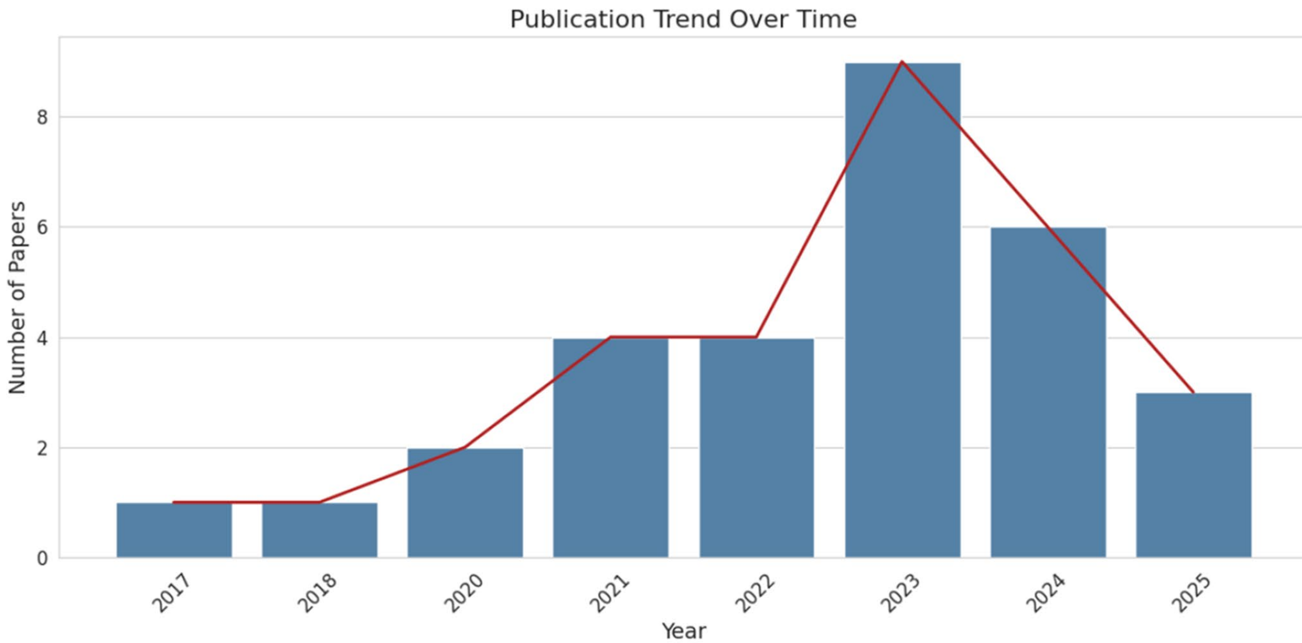


Fig. 6 Publication trend within the study domain DL for AC

with its 13 spectral bands, provides high-resolution optical imagery suitable for land surface monitoring and AC. MODIS, known for its daily global coverage, is instrumental in large-scale AC applications. With its long historical archive, Landsat allows researchers to study atmospheric effects over extended periods, making it valuable for long-term trend analysis. In contrast, hyperspectral datasets, including AVIRIS [38] and Hyperion [39], capture detailed spectral information across hundreds of narrow bands, enabling precise SR retrieval. The high dimensionality of these datasets poses computational challenges for AC.

Access to these datasets is facilitated through publicly available repositories. Sentinel-2 data can be retrieved from the Copernicus Open Access Hub[4], while MODIS imagery is accessible via NASAs Earthdata portal[5]. Landsat datasets are available through USGSs Earth Explorer platform[6]. Hyperspectral datasets such as AVIRIS can be obtained from NASAs Airborne Data Systems, Hyperion imagery is archived within the USGS Earth Explorer system. These platforms provide researchers with a wealth of freely available satellite imagery, ensuring reproducibility and enabling large-scale model training and validation. Table 1 comprehensively summarises various datasets commonly used in DL-based AC. The table captures the diversity of data sources across key dimensions such as spectral resolution (multispectral vs. Hyperspectral), revisit frequency, spatial resolution, and data accessibility. For instance, platforms like Landsat-8 and Sentinel-2 offer globally available, high-resolution multispectral imagery. They are frequently used due to their consistent temporal coverage and ARD availability. On the other hand, hyperspectral missions such as AVIRIS and Hyperion provide detailed spectral profiles that are valuable for fine-scale surface characterisation. Sensors like MODIS and Himawari-8 provide high temporal resolution suitable for studying atmospheric dynamics, while ocean colour missions like GOCI and HY-1C cater specifically to marine and coastal studies.

3.2 Simulated Datasets

In DL-based AC, the availability of high-quality training data that includes both TOA reflectance and corresponding SR is critical. However, acquiring large-scale, co-located datasets of TOA and SR under diverse atmospheric, seasonal, and geographical conditions is inherently challenging due to logistical constraints, sensor limitations, and the impracticality of exhaustive ground-truth collection. Several studies [47–51] have turned to generating simulated datasets using physics-based radiative transfer models to overcome these challenges. These simulations aim to mimic the interaction of solar radiation with atmospheric constituents and surface features, thereby enabling the generation of large, diverse, and controllable datasets for training robust DL models.

Simulated datasets are typically created by coupling spectral libraries or SR measurements with radiative

[4] https://www.copernicus.eu/en/access-data/conventional-data-access-hubs

[5] https://www.earthdata.nasa.gov/

[6] https://earthexplorer.usgs.gov/

Table 1 Summary of satellite datasets commonly used for atmospheric correction

Sensor / Mission	Type	Spectral Bands	Spatial Resolution	Temporal Resolution	Data Availability	Products Available	Geographic Coverage	Type of Coverage
Landsat 8 (OLI) [37]	Multispectral	11 bands (VIS, NIR, SWIR, TIR)	15 m (pan), 30 m	16 days	Free (USGS EarthExplorer)	TOA (Level-1), BOA (Level-2), Cloud mask	Global	Land
Sentinel-2 (MSI) [35]	Multispectral	13 bands (VIS, NIR, SWIR)	10 m, 20 m, 60 m	5 days	Free (Copernicus Open Access Hub)	TOA (Level-1C), BOA (Level-2A), Cloud Mask	Global	Land
Sentinel-3 (OLCI) [40]	Multispectral	21	300 m	Almost Daily	Free (Copernicus)	TOA, BOA, Ocean colour	Global	Ocean
MODIS Terra/Aqua [36]	Multispectral	36 bands (VIS–TIR)	250 m, 500 m, 1 km	1–2 days	Free (LAADS DAAC)	Ocean colour, TOA, BOA, Cloud Mask, AOD, NDVI	Global	Land, Ocean, Atmosphere
Hyperion (EO-1) [39]	Hyperspectral	220 bands (VIS–SWIR)	30 m	16 Days	Archived (USGS EarthExplorer)	TOA, BOA	Global	Land
AVIRIS (Airborne) [38]	Hyperspectral	224 bands (400–2500 nm)	4–20 m (depends on altitude)	On-demand (campaign-based)	Free (JPL)	TOA, BOA	Specific regions (USA, airborne)	Land
AISA Eagle [41]	Hyperspectral (Airborne)	64-288 (400–1000 nm)	1–5 m (airborne)	On-demand (campaign-based)	Commercial	TOA, BOA	Specific airborne missions (regional)	Land
Himawari-8 (AHI) [42]	Multispectral	16 bands (VIS–TIR)	0.5–2 km	10 min	Free (JMA/NASA)	TOA, BOA, Cloud Mask, AOD	East Asia, Australia, Western Pacific	Land
GOCI [43]	Multispectral	8 (VIS-NIR)	500 m	1 hour	Free (KIOST/NASA)	Ocean colour, AOD, TOA	Korean Peninsula and surrounding	Ocean
PERUSAT-1 [44]	Multispectral	4 bands (VIS, NIR), Pan	2 m (pan), 10 m (MS)	5–15 days	Commercial	TOA	Peru (South America)	Land
HY-1C (CZI) [45]	Multispectral	4 bands (VIS-NIR)	50 m	3 days	Limited (Chinese space agency)	TOA, BOA Ocean colour	China coastal regions and nearby seas	Ocean and Coastal Areas
LISS-3 (IRS-1C/1D) [46]	Multispectral	4 bands (VIS, NIR, SWIR)	23.5 m	24 days	Free (Govt. of India/ USGS EarthExplorer)	TOA, BOA, NDVI	Global	Land

transfer models such as MODTRAN or 6 SV. A set of well-defined input parameters must be specified to generate simulated datasets to drive the radiative transfer models. These typically include atmospheric profiles (such as AOD, water vapour, and ozone concentration), solar and sensor geometry (including solar zenith, sensor zenith, and relative azimuth angles), surface characteristics (such as land cover type or water properties), and viewing conditions (e.g., altitude and surface pressure). For instance, some works generate synthetic TOA-SR pairs by sampling reflectance from vegetation libraries [47] and propagating them through a modelled atmosphere under multiple viewing and illumination geometries [49]. Authors in [50, 51] have used SeaDAS software to generate SR reflectances from the TOA reflectances under varying atmospheric and water conditions. Others simulate the entire image acquisition pipeline, incorporating sensor-specific response functions to ensure that the synthetic data closely resembles observations from hyperspectral or multispectral instruments [48].

3.3 Ground Data for Validation

RadCalNet [52] and AERONET-OC [53] are the primary datasets for validating DL model-predicted SR values. These datasets provide ground-truth SR measurements obtained through direct in situ observations rather than simulations or physics-based AC models, ensuring their authenticity. RadCalNet primarily focuses on land-based locations, offering well-calibrated SR data from radiometrically stable test sites essential for evaluating satellite-derived products over terrestrial regions. In contrast, AERONET-OC provides SR measurements over open and coastal waters, making it a crucial dataset for validating ocean colour retrievals and assessing AC models in aquatic environments. In addition to these global networks, researchers often conduct field campaigns to collect in situ data using spectroradiometers.

3.3.1 RadCalnet

RadCalNet [52] is an initiative of the Working Group on Calibration and Validation of the Committee on Earth Observation Satellites (CEOS) [6]. This service provides satellite operators with TOA reflectance measurements, which are critical for post-launch radiometric calibration and optical imaging sensor data validation. The network consists of multiple well-characterised test sites carefully selected for their stable and uniform surface properties, ensuring reliable ground truth (GT) data for satellite calibration. The primary RadCalNet sites include Railroad Valley Playa in Nevada, Gobabeb in Namibia, La Crau in France, and Baotou in China, each offering unique spectral and environmental characteristics for rigorous calibration and validation activities.

Railroad Valley Playa in Nevada, USA, is a dry lake bed known for its highly homogeneous and stable SR, making it an ideal site for optical sensor calibration. The arid environment minimises atmospheric variability, leading to consistent observations. Reflectance values at this site represent a 1 km × 1 km area centred at Lat: 38.497° and Lon: −115.69°. La Crau in France is a semi-arid grassland featuring a well-characterised mix of vegetation and soil reflectance, making it valuable for multispectral and hyperspectral sensor validation. The reflectance spectra at this site represent a disk of 30 m radius centred at Lat: 43.5588° and Lon: 4.8641°. Gobabeb in Namibia, situated in the Namib Desert, is characterised by expansive dunes that provide a bright and stable reflective surface. The low vegetation cover and predictable atmospheric conditions make it suitable for radiometric calibration. Reflectance spectra here correspond to a disk of 30 m radius centred at Lat: −23.6002° and Lon: 15.11956°. The Baotou site in Inner Mongolia, China (Lat: 40.8514, Lon: 109.6291), spans approximately 300 km^2 and features diverse land covers including lake, sand, bare soil, and various crops. It also includes a permanent greyscale target composed of two white, one grey, and one black gravel blocks, each measuring 48 × 48 meters, required for calibration. Figure 7 illustrates a Landsat 8 satellite tile encompassing the RadCalNet calibration sites, while Fig. 8 presents ground-level images of the actual site locations.

The RadCalNet service provides access to TOA and bottom-of-atmosphere reflectance data. These measurements, accompanied by associated uncertainties, are made available at a spectral sampling interval of 10 nm, covering wavelengths from 380 nm to 2500 nm, and are updated every 30 minutes. Each RadCalNet site has automated ground instruments that continuously measure SR alongside local environmental and atmospheric conditions. Data acquisition occurs every 30 minutes between 09:00 and 15:00 local time. The collected measurements are processed using a standardised methodology within a central processing system, ensuring uniformity in TOA reflectance data across all sites.

Thus, RadCalNet is an essential reference for verifying physics- and DL-based AC algorithms. For an accurate and fair evaluation of AC models, satellite imagery and RadCalNet data must be closely matched in temporal acquisition. Since atmospheric conditions can change rapiDLy due to variations in aerosols, water vapour, and cloud cover, using RadCalNet observations acquired near the satellite overpass time is crucial to minimise discrepancies.

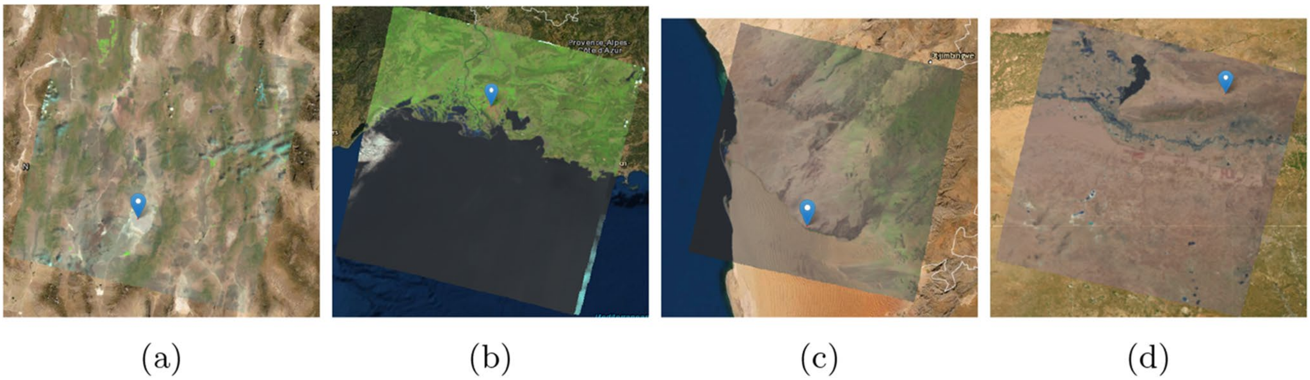


Fig. 7 Landsat-8 oli tile of (**a**): railroad-USA (**b**): LaCrau-france (**c**): gobabeb-namibia and (**d**): Bataou-China

Fig. 8 Site images of (**a**): railroad-USA, (**b**): LaCrau-france, (**c**): gobabeb-namibia, and (**d**): bataou-China [52]

3.3.2 AERONET-OC

AERONET-Ocean Colour [53] is an extension of the Aerosol Robotic Network (AERONET), specifically designed to provide continuous, high-quality measurements of aerosol optical properties and water-leaving radiance to support ocean colour remote sensing applications. Managed by NASA in collaboration with international partners, AERONET-OC consists of a globally distributed network of autonomous sun photometers deployed at coastal and open-ocean sites. These instruments measure spectral radiance and AOD across multiple wavelengths, critical in calibrating and validating satellite-based ocean colour retrievals and AC algorithms.

The primary objective of AERONET-OC is to provide standardised in situ ocean colour observations that support the development and refinement of remote sensing algorithms. The network's strategic deployment across diverse marine environments ensures comprehensive coverage of varying optical water types. Key sites include Venice (Italy), Gustav Dalen Tower (Sweden), WaveCIS (USA), and Gloria (Spain), each representing different bio-optical conditions. Figure 9 shows the locations of different sites worldwide.

AERONET-OC instruments operate in direct sun and sky radiance measurement modes, allowing for precise quantification of aerosol optical properties and their impact on ocean colour observations. The network provides essential parameters such as spectral AOD, Ångström exponent, and aerosol size distribution, which are fundamental for AC in ocean colour remote sensing. Additionally, AERONET-OC measures normalised water-leaving radiance, a key variable used for validating remote sensing reflectance (RSR) obtained from satellite-based sensors, including MODIS, VIIRS, Sentinel-3 OLCI, and Landsat-8/9 OLI. AERONET-OC plays a pivotal role in evaluating the performance of physics- and DL-based AC models. Since ocean colour remote sensing is highly sensitive to atmospheric conditions, accurate AC is essential to retrieve precise water-leaving reflectance. The spectral and temporal consistency of AERONET-OC data allows for robust validation of AC models by providing ground-truth observations for comparison with satellite-derived reflectance values. For fair model assessments, it is crucial to use AERONET-OC measurements acquired as close as possible to the satellite overpass time, minimising discrepancies caused by rapid atmospheric variations and transient oceanographic processes.

One of the key challenges in using AERONET-OC data for satellite validation arises from differences in spectral bands between satellite sensors and ground-based instruments. Each AERONET-OC station is equipped with different radiometers, leading to variations in sensor characteristics, such as spectral response functions and measurement precision. To reconcile these band differences, a band-shifting

Fig. 9 AERONET-OC locations on world map [54]

technique is typically required. However, this approach depends on ancillary information, such as water absorption and scattering coefficients, which are not always available at the necessary spatial and temporal resolutions. Additionally, satellite-derived estimates of these optical properties in coastal waters are often associated with significant uncertainties due to complex bio-optical conditions, suspended sediments, and variable atmospheric contributions.

To mitigate these challenges, an alternative approach involves applying interpolation techniques to correct for spectral band differences between in situ and satellite observations. While interpolation may not be as accurate as physics-based spectral adjustments that require extensive ancillary data, it provides a practical and widely applicable solution for harmonising spectral measurements. By ensuring better spectral alignment between satellite and ground-based measurements, interpolation techniques improve the consistency of ocean colour validation efforts, reducing potential discrepancies introduced by sensor-specific spectral differences.

3.3.3 Field Campaigns

Publicly available datasets such as RadCalNet and AERONET-OC offer standardised, well-calibrated observations from ground sites and ocean colour monitoring stations, respectively. However, these resources are limited in spatial and temporal coverage.

To supplement such data, researchers often conduct dedicated field campaigns [55, 56], using various portable and fixed instruments to collect in situ measurements. Spectroradiometers are commonly used for land surfaces, either handheld or mounted on tripods. During satellite overpasses, these devices measure upwelling radiance from the Earth's surface and downwelling irradiance from the atmosphere. SR is then derived as the ratio of these two measurements. To ensure consistency and minimise noise, multiple readings are typically taken over homogeneous land cover types and averaged.

For aquatic environments, specialised instruments such as the DALEC system [57], shipborne radiometers [58], and the Marine Optical Buoy (MOBY) [59] are employed. DALEC and shipborne sensors capture radiometric measurements from the water surface, often complemented by atmospheric parameters like aerosol and water vapour content. MOBY, in particular, is a moored buoy equipped with radiometers positioned above and below the water surface, allowing it to measure both water-leaving radiance and incident sunlight. These measurements are routinely calibrated and temporally synchronised with satellite acquisitions to serve as reliable validation data.

3.4 Evaluation Metrics

To validate the performance of the DL model, it is essential to compare the model-predicted SR with reference SR values. These reference SR values may come from satellite-based test datasets, physically simulated datasets generated through radiative transfer models, or field-based measurements obtained via spectroradiometers or standardised networks like RadCalNet and AERONET-OC. Several quantitative metrics are commonly used to assess model performance in this context, providing insights into prediction accuracy, structural consistency, and overall reliability of the DL model. Some important performance metrics are as follows:

- Root Mean Square Error: RMSE, as defined in Eq. 2, measures the extent to which the predicted SR values deviate from their corresponding reference or ground-truth values. It is a commonly used metric for evaluating the accuracy of DL models in AC tasks. RMSE computes the square root of the average of squared differences between predicted and observed values, providing a single scalar value that reflects the magnitude of prediction errors in the same units as the original measurements. Its interpretability and sensitivity to large errors make it a preferred choice for performance evaluation and model comparison in various applications.

$$RMSE_{\lambda} = \sqrt{\frac{\sum_{i=1}^{n_{\lambda}} (\Delta\rho_{i,\lambda})^2}{n_{\lambda}}} \quad (2)$$

  $RMSE_{\lambda}$ is the RMSE value between the SR values predicted by the model and those derived from satellite for band λ, n_{λ} is the number of pixels in band λ, $\Delta\rho_{i,\lambda}$ is the difference of model predicted SR values and satellite predicted SR values defined as Eq. 3.

$$\Delta\rho_{i,\lambda} = \hat{\rho_{i,\lambda}} - \rho_{i,\lambda} \quad (3)$$

  where, $\hat{\rho_{i,\lambda}}$ is the model predicted SR and $\rho_{i,\lambda}$ is the reference SR for pixel i in band λ.

- Correlation Coefficient (r): The correlation coefficient is a widely used statistical metric that quantifies the strength and direction of a linear relationship between two continuous variables—in this case, the predicted SR values from a DL model and the corresponding ground-truth SR values. It ranges from −1 to 1, where a value close to 1 indicates a strong positive correlation, suggesting that the predicted values closely follow the trend of the reference data. A coefficient near 0 implies no linear relationship, while values approaching −1 signify a strong inverse correlation. By evaluating the correlation coefficient, we can assess how effectively the model preserves the underlying spatial and spectral distribution of SR.

$$r_{\lambda} = \frac{\sum_{i=1}^{n} (\rho_{i,\lambda} - \mu_{\lambda})(\hat{\rho_{i,\lambda}} - \hat{\mu_{\lambda}})}{\sqrt{\sum_{i=1}^{n} (\rho_{i,\lambda} - \mu_{\lambda})^2}\sqrt{\sum_{i=1}^{n} (\hat{\rho_{i,\lambda}} - \hat{\mu_{\lambda}})^2}} \quad (4)$$

  r_{λ} is the correlation coefficient between the satellite BOA image and the model-predicted image for band λ. $\rho_{i,\lambda}$ is the satellite predicted SR, and $\hat{\rho_{i,\lambda}}$ is the predicted SR of the model for pixel i in band λ. μ_{λ} and $\hat{\mu_{\lambda}}$ are the mean SR of satellite BOA image and model predicted image for band λ, respectively.

- Mean Absolute Percentage Error (MAPE): A widely used evaluation metric that quantifies the average percentage difference between predicted and actual values. It provides an intuitive measure of prediction accuracy by expressing the error as a percentage of the true values, as shown in Eq. 5. MAPE is especially useful in applications where understanding relative errors is more meaningful than absolute deviations.

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\rho_i - \hat{\rho}_i}{\rho_i}\right| \times 100 \quad (5)$$

  where ρ_i represents the actual values, $\hat{\rho}_i$ represents the predicted values, and n is the total number of observations. MAPE is advantageous due to its interpretability, as it expresses error in percentage terms, making it easier to compare across datasets with different scales. However, it has limitations, particularly when dealing with values close to zero, as the division by small numbers can lead to excessively high error percentages, skewing the overall evaluation. This sensitivity makes MAPE less suitable for datasets with many near-zero observations.

4 Deep Learning for Atmospheric Correction

4.1 Training Strategies and DL Models

DL-based AC models are trained using two primary strategies: pixel-level and image-level processing, each offering distinct advantages and trade-offs regarding spectral accuracy, spatial coherence, and computational efficiency.

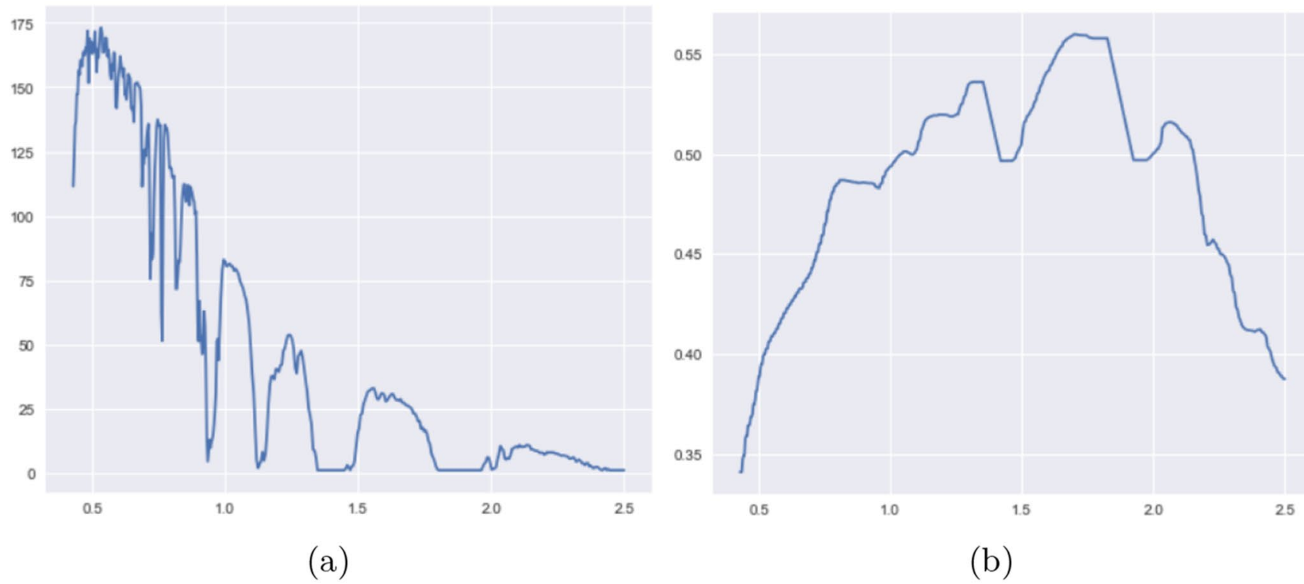


Fig. 10 (**a**): TOA reflectance and (**b**): BOA reflectance − 500 nm to 2500 nm wavelength

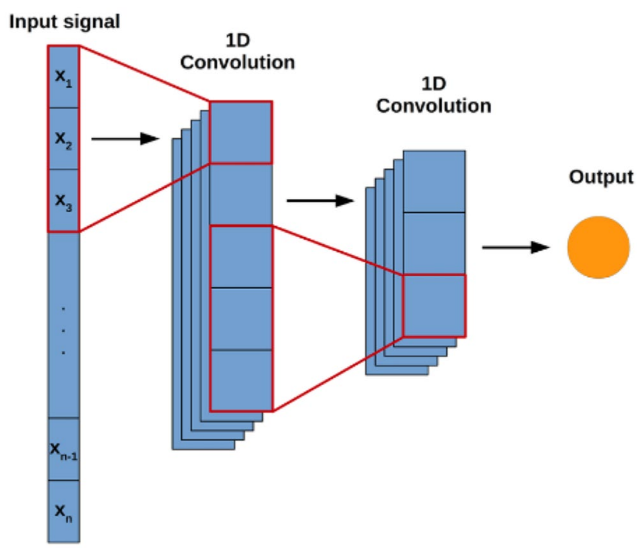


Fig. 11 A 1D CNN architecture [60]

4.1.1 Pixel Level Processing

Pixel-level processing involves training DL models to learn the relationship between TOA and BOA reflectance at an individual pixel level. Figure 10 shows the TOA and BOA reflectance plot. Each pixel is treated independently, making this approach computationally efficient and suitable for large-scale applications such as global AC of satellite imagery.

Common pixel-based DL architectures for AC include:

- Time-dependent Encoder–Decoder Networks: Time-dependent Encoder–Decoder Networks [47, 48] extend conventional encoder–decoder architectures by incorporating temporal metadata—such as acquisition date, time of day, and day-of-year—into the learning process. This temporal conditioning allows the network to model diurnal and seasonal atmospheric variability, which is crucial because atmospheric properties and illumination geometry can vary significantly over time, even for the same geographic location.
- One-Dimensional Convolutional Neural Networks (1D CNNs): 1D CNNs process hyperspectral or multispectral pixel vectors as one-dimensional sequences along the spectral dimension as shown in Fig. 11. The convolution operation for a 1D CNN is expressed as:

$$y_i = \sum_{j=0}^{k-1} w_j \cdot x_{i+j}, \qquad (6)$$

where x is the input sequence, w_j is the kernel weight, and y_i is the output feature map. Each convolutional filter operates across contiguous spectral bands, enabling the network to capture local spectral dependencies arising from surface reflectance characteristics and atmospheric absorption features. They are particularly effective for TOA-to-BOA transformations in optically complex environments, such as inland and coastal waters or vegetation-rich areas [61, 62].
- Autoencoders (AEs): Autoencoders (AEs) are unsupervised neural architectures that learn a compact latent representation of input data by minimising reconstruction error between the input and output. AEs can be trained to map TOA reflectance to SR by implicitly

learning to filter out atmospheric distortions [63]. The encoder compresses the spectral–spatial information into a latent code that ideally captures only the intrinsic surface properties, while the decoder reconstructs the corrected SR from this representation. Denoising Autoencoders (DAEs) extend this principle by intentionally corrupting the input—e.g., adding synthetic noise or atmospheric artefacts—and training the network to reconstruct the SR.

- Fully Connected Feedforward Neural Networks (FCNNs): FCNNs process spectral data without convolutional operations. Each neuron in a layer is connected to every neuron in the subsequent layer, enabling the network to capture global dependencies across the spectrum. These models often estimate atmospheric properties such as path radiance, transmission coefficients, and scattering effects [55, 64].
- Multilayer Perceptrons (MLPs): MLPS are a specific type of FCNN with multiple hidden layers of neurons. Each neuron computes a weighted sum of its inputs, followed by a nonlinear activation function. The layer-wise transformation is given by:

$$\mathbf{h}^{(l)} = f(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \tag{7}$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weights and biases, and f is a non-linear activation such as ReLU. A schematic diagram of MLP is shown in Fig. 12. MLPs are widely applied in pixel-level AC tasks due to their flexibility and ability to learn complex non-linear mappings from TOA reflectance to BOA reflectance [50, 51, 58, 62, 65–68].

4.1.2 Image Level Processing

On the contrary, image-level processing involves training models on entire satellite images or large patches, enabling them to capture spatial dependencies and regional atmospheric variations. CNNs and transformer-based architectures are commonly employed in image-level processing to extract spatial features and improve the robustness of AC models.

Common image-level DL architectures for AC include:

- Two-Dimensional Convolutional Neural Networks (2D CNNs): 2D CNNS are widely used in computer vision and image analysis tasks. 2D convolutional kernels are applied to extract spatial features from images. Each kernel scans over the input matrix to detect spatial patterns such as edges, textures, and shapes. The convolution is followed by non-linear activations and pooling operations that reduce spatial resolution and introduce invariance to local distortions. Prominent architectures include AlexNet [69], VGGNet [70], and ResNet [71], which are commonly used in segmentation [72], classification [73], and object detection [74] tasks in Earth observation.

  These networks extract spatial correlations between neighboring pixels within each spectral band, reducing

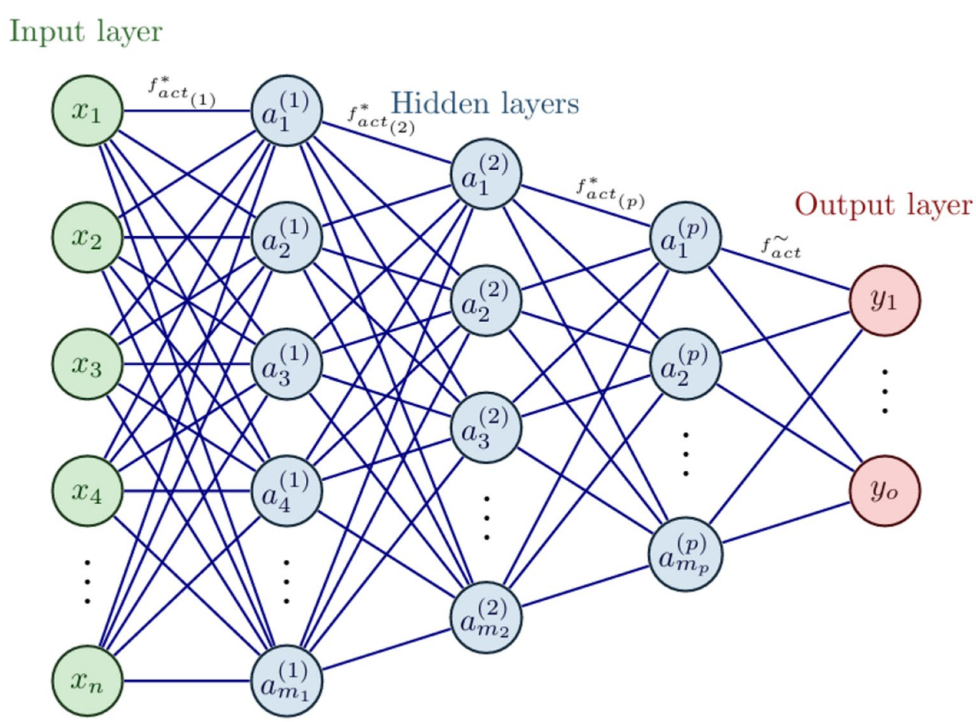


Fig. 12 A feedforward neural network or multilayer perceptron architecture [64]

atmospheric noise and haze in multispectral imagery [49, 56, 75–77]. They are efficient for applications with the primary focus on spatial feature extraction. At the same time, spectral information is hanDLed implicitly through multi-channel input representations, where each spectral band is treated as an image channel. However, these networks may not fully exploit inter-band spectral relationships, which can limit performance in hyperspectral AC unless combined with spectral processing modules or hybrid spatial–spectral architectures.

- Three-Dimensional Convolutional Neural Networks (3D CNNs): Extending convolution into the spectral dimension, 3D CNNs jointly learn spatial and spectral dependencies. This makes them highly effective for hyperspectral AC tasks, where fine-grained spectral–spatial interactions are critical for accurate surface reflectance retrieval [49, 56, 75–77]. However, their increased computational and memory demands can hinder scalability for global or real-time applications, motivating ongoing research into lightweight and hybrid spatial–spectral designs.
- Discrete–Continuous CNN (DC-CNN): DC-CNNs are designed to bridge the gap between discrete spectral sampling inherent in multispectral or hyperspectral sensors and the need for continuous spectral representation in AC tasks. While conventional CNNs process only the measured spectral bands, DC-CNNs incorporate a continuous spectral reconstruction module that models reflectance values between observed wavelengths [78]. This design enables the network to preserve band-to-band continuity, which is particularly important for hyperspectral datasets.
- Encoder-Decoder Architecture: The encoder-decoder architecture transforms input data into a lower-dimensional latent representation and reconstructs a meaningful output. The encoder compresses the input into a compact feature space. At the same time, the decoder expands this representation to generate the output as shown in Fig. 13. Formally, the encoder $\mathcal{E}$ maps the input $\mathbf{x}$ to a latent vector $\mathbf{z}$. The decoder $\mathcal{D}$ reconstructs $\hat{\mathbf{y}} = \mathcal{D}(\mathcal{E}(\mathbf{x}))$. U-Net [80] and SegNet [81] are prominent examples applied to semantic segmentation and land cover mapping.
U-Net, originally proposed for biomedical image segmentation [79], is now extensively used in AC tasks. Its encoder progressively downsamples the input image, extracting increasingly abstract spatial features. At the same time, the decoder upsamples these representations to the original resolution, reconstructing spatially resolved outputs such as SR. A defining characteristic of U-Net is skip connections that link corresponding encoder and decoder layers, preserving high-resolution spatial information and improving spatial–spectral fidelity during reconstruction. R2U-Net extends the standard U-Net by incorporating residual learning and recurrent convolutional layers into its encoder–decoder framework. Residual blocks facilitate the learning of complex mappings by allowing the network to refine corrections relative to its input, improving convergence stability and mitigating vanishing gradient issues. R2U-Net is particularly effective for correcting heterogeneous land surfaces influenced by variable atmospheric effects [82].
- Swin Transformer: The Swin Transformer [83] is a hierarchical vision transformer architecture that computes

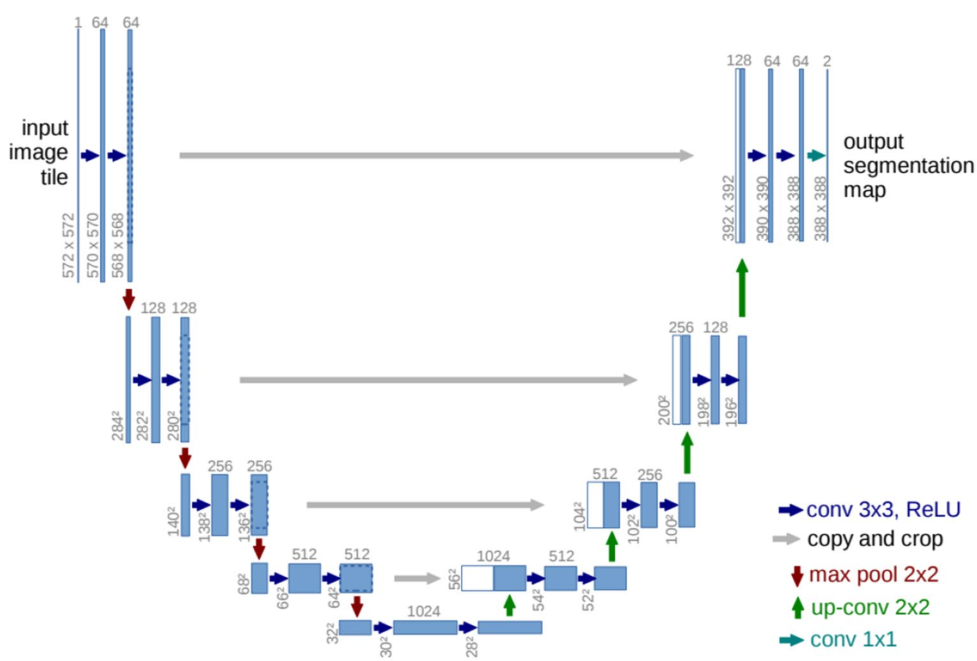


Fig. 13 U-net encoder-decoder DL model [79]

self-attention within non-overlapping local windows, which are shifted between successive layers to enable cross-window connections. This design significantly reduces computational complexity from quadratic to linear with respect to image size, making it feasible for high-resolution remote sensing imagery. The Swin Transformer captures local and global spatial dependencies by progressively merging image patches and increasing the receptive field.
In AC, the Swin Transformer can effectively refine spectral–spatial features by leveraging its multi-scale representation learning, improving robustness against haze, scattering, and illumination variations. In the RCFormer framework [84], the Swin Transformer is integrated with parallel convolutional branches to fuse transformer-derived global context with CNN-extracted local details. This hybrid design enhances the model's ability to recover fine-grained SR structures while maintaining global radiometric consistency. Such architectures are particularly beneficial for UAV-based multispectral imagery, where variable atmospheric conditions and high spatial resolution demand a balanced capture of long-range dependencies and localized corrections.

- SAAC-Net (Seasonal Atmospheric Adjustment Correction Network): A CNN-based architecture with residual blocks, SAAC-Net [30] incorporates a synthetic "season band" to model intra-annual variations in atmospheric effects. Residual learning enables fine-grained correction without distorting essential spectral signatures. It is suitable for dynamic environments where atmospheric conditions vary over time.

Both pixel-level and image-level processing strategies have inherent trade-offs. Pixel-based approaches offer high spectral accuracy and computational efficiency, making them suitable for large-scale, real-time AC applications. However, their inability to incorporate spatial dependencies results in inconsistencies, particularly in complex terrains. Image-based approaches, in contrast, preserve spatial coherence by leveraging regional atmospheric variations but are computationally expensive and require larger, more diverse training datasets to ensure generalisation across different atmospheric conditions.

Future advancements in DL-based AC will likely focus on developing hybrid models that integrate pixel-level and image-level processing, leveraging the strengths of both approaches. These models will aim to balance spectral fidelity, spatial consistency, and computational efficiency, ensuring scalable, real-time AC solutions for large-scale Earth observation applications.

4.2 Deep Learning Approaches for Atmospheric Correction: Physics Aware and Physics Agnostic

The development of DL-based AC models has led to two distinct approaches: physics-aware and physics-agnostic methods. These approaches differ in how they incorporate atmospheric physics into learning. Physics-aware models utilise auxiliary inputs such as atmospheric and geometric parameters and TOA reflectance during learning. In contrast, physics-agnostic models rely solely on TOA reflectance, learning direct mappings to bottom-of-atmosphere or SR without explicitly requiring physical input variables.

The bar chart in Fig. 14 compares physics-aware and physics-agnostic DL models based on their target environments: vegetation, coastal zone, land, and ocean. Most AC approaches aimed at coastal zones and oceanic regions adopt physics-aware models. For land-based correction tasks, both types of approaches have been explored. Interestingly, only one study in the reviewed literature specifically targets vegetation and employs a physics-agnostic model.

Tables 2, 3, and 4 provide a consolidated overview of DL-based AC methods categorised by their modelling approach. These tables include essential details such as the datasets employed, the underlying physics models (if any), target environments (e.g., land, ocean, vegetation), DL architectures, and reported performance outcomes. For consistency and comparability, the results included in these tables represent the best-performing configurations among various experiments conducted in each study. Specifically, the performance metrics presented—RMSE and correlation coefficient (r)—are the average values reported across different spectral bands.

We also indicate whether a model qualifies as end-to-end. A DL model is considered end-to-end if it directly outputs SR values from TOA inputs. Conversely, suppose that the model estimates intermediate radiative transfer parameters that require further post-processing to compute SR. In that case, it is not classified as end-to-end. For instance, the study by [48] proposes a time-dependent deep neural network (DNN) for AC of shortwave hyperspectral remote sensing data. The model uses multi-angle hyperspectral data to estimate six radiative transfer components—including atmospheric transmission, path radiance, downwelling radiance, and solar scattering contributions. These intermediate outputs are then used to compute SR values, making the model non-end-to-end. Another study [55] presents an ML-based method to accelerate AC for Resourcesat-2a (R2A) LISS-3 multispectral imagery. The approach simplifies the 6S RTE into a linearised form:

$$\rho_{BOA} = \frac{(\rho_{TOA} - a)}{b} \tag{8}$$

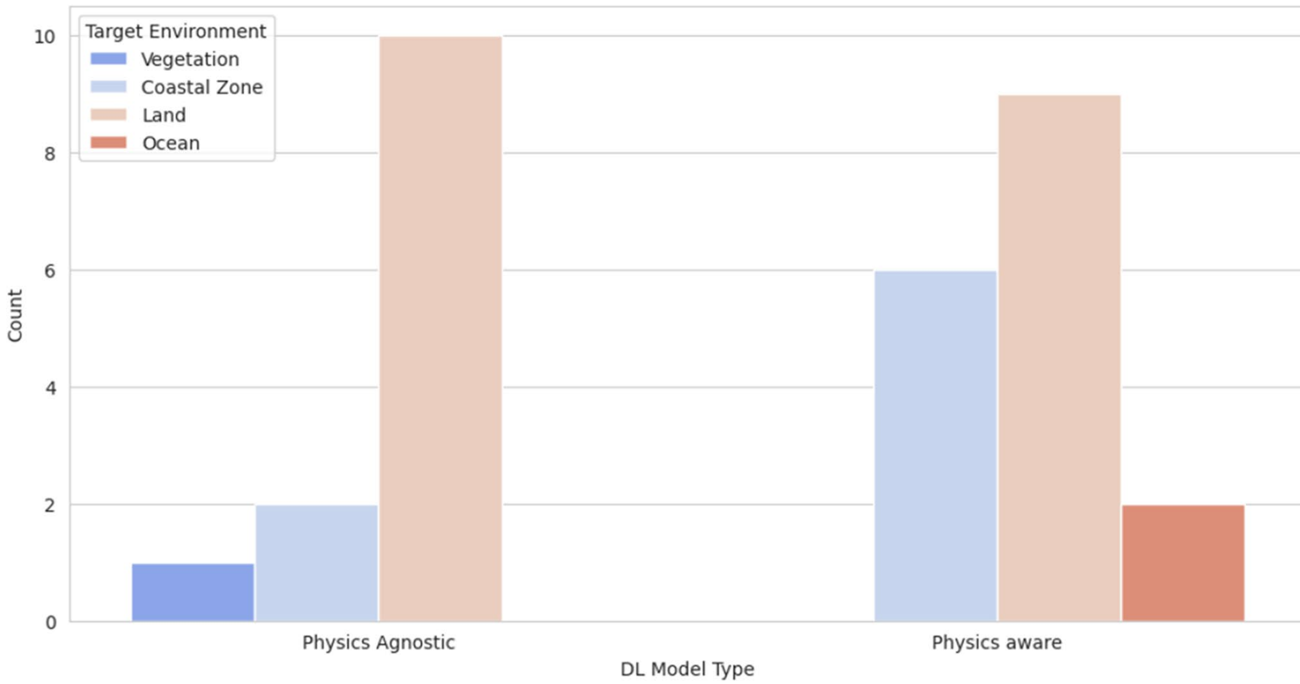


Fig. 14 Number of DL models for different target environments - vegetation, coastal zone, land, and ocean

AC coefficients *a* and *b* are predicted by a fully connected neural network (FCNN). The model is trained on inputs such as water vapour, aerosol optical thickness (AOT), ozone, azimuth angle, elevation angle, and altitude. Once these coefficients are estimated, they are substituted into the simplified equation to compute BOA reflectance efficiently.

4.2.1 Physics-Aware DL Models

Physics-aware DL models enhance the accuracy of AC by integrating atmospheric and geometric parameters alongside TOA radiance. These parameters typically include geometric parameters, atmospheric properties, and radiative transfer-related variables.

- Geometric Parameters: Geometric parameters, such as solar zenith angle (SZA), view zenith angle (VZA), and relative azimuth angle (RAA), define the illumination and viewing geometry.
  - SZA influences surface illumination and atmospheric path length, with higher angles (e.g., sunrise/sunset) causing greater scattering.
  - VZA affects atmospheric path length and introduces Bidirectional Reflectance Distribution Function (BRDF) effects, especially over heterogeneous surfaces like vegetation or urban areas.
  - RAA, representing the angular difference between solar and sensor azimuths, impacts SR variability under different observation conditions.

Tables 2 and 3 summarises different physics-aware DL approaches. Most physics-aware DL-based AC models use SZA, VZA, and RAA ([50, 56, 58, 59, 62, 68, 75, 78, 86, 90]. In addition to these parameters, authors have explored adding a few more parameters. For instance, authors in [61] incorporated canopy properties (LAI, LIDF, Hspot) and leaf traits (chlorophyll, leaf structure parameters) in addition to geometric angles. Their approach aimed to simulate the PROSAIL model, one of the most widely used RTMs in remote sensing, which combines the PROSPECT (leaf model) and SAIL (canopy model) to define directional reflectance as a function of leaf properties, canopy structure, sun-surface-sensor geometry, and soil reflectance. Authors in [49] used eight different elevation angles (ranging from 30° to 90° in 5° increments) to capture variations in atmospheric transmission, scattering, and upwelling/downwelling radiation, allowing the model to learn how these factors influence TOA radiance from multiple viewing perspectives.

- Atmospheric Properties: Atmospheric parameters such as aerosol optical depth (AOD) and columnar water vapour (CWV) directly influence TOA radiance and are critical for AC.

Table 2 Summary of physics-aware DL models for AC (part-1)

Paper	Auxiliary Inputs Used (Other than TOA Radiance)	Physics Model	Dataset Used	Multispectral/ Hyperspectral	Spatial Processing (Pixel/ Image)	Target Environment	Model Used	End-to-End?	Result	Validation using Ground Data?
[61]	SZA, VZA, RAA, Canopy Properties (LAI, LIDF, Hspot), Leaf traits (Chlorophyll, leaf structure parameter)	PROSAIL + 6S	Synthetic Ocean and Land colour Imager (OLCI)	Multispectral	Pixel	Land	1D CNN	No	RMSE: 0.0026, r: 0.98	No
[75]	SZA, VZA, RAA	HY-1 CZI	HY-1 CZI, Landsat-8 OLI and in situ from AERONET-OC	Multispectral	Image	Coastal Zone	2D and 3D convolution	Yes	Test Results RMSE: 0.0006, r: 0.96; In-Situ Results RMSE: 0.0003, r: 0.99	AERONET-OC
[78]	SZA, RAA	MAIAC	Himawari-8	Multispectral	Image	Land	Full Bayesian architectures conditioned to learn a discrete-continuous distribution	Yes	CRMSE: 0.0206, r: 0.90	No
[58]	SZA, SnZA, RAA	HY-1 CZI	HY-1 CZI + Landsat	Multispectral	Pixel	Coastal Zone	Multilayer Neural Network	Yes	Test Results RMSE: 0.0003, r: 0.97; In-Situ Results RMSE: 0.006, r: 0.72	AERONET-OC Shipborne Measurement
[63]	Total flux, Diffuse transmittance, Direct transmittance, Spherical albedo, Path radiance	MODTRAN 6	AISA Eagle Sensor + Ground	Hyperspectral	Pixel	Land	AutoEncoders	Yes	RMSE: 0.00018, r: 0.74	No
[68]	SZA, SnZA, RAA	MODIS	MODIS aqua collection 6	Multispectral	Pixel	Coastal Zone	Multilayer Neural Network	No	In-Situ Results APD: 3.15%, r: 0.99	AERONET-OC
[49]	8 geometric angles	MODTRAN	Synthetic generated using MODTRAN software	Hyperspectral	Pixel	Land	CNN	No	RMSE: 0.02	No
[55]	Water vapor, AOT, ozone, azimuth angle, elevation angle, altitude	6S	LISS-3	Multispectral	Pixel	Land	Fully connected feed forward NN	Yes	Test Results RMSE: 0.00022, r: 0.97; In-Situ Results RMSE: 0.0008	Spectro-Radiometer and Sun-Photometer

- AOD is a critical parameter that quantifies the concentration of aerosols in the atmosphere, directly influencing TOA radiance. Higher AOD values increase scattering and absorption, particularly in the blue and shortwave-infrared (SWIR) bands [91].
- Water vapour content, another essential factor, strongly absorbs radiation in the near-infrared (NIR) and infrared regions [91]. Errors in reflectance retrieval due to water vapour variations are influenced by sensor sensitivity [92], spectral absorption features [93], and the accuracy of radiative transfer models [94].
- Ozone concentration (O3) significantly impacts UV and visible bands by absorbing solar radiation [95].

Authors in [64] integrated AOD, CWV, ozone (O3), Nadir Bidirectional Reflectance Distribution Function Adjusted Reflectance (NBAR), Digital Elevation Model (DEM), terrain characteristics (aspect, slope, shadows), and sun–sensor acquisition geometry values as inputs to their AC model. These additional features allowed the model to capture the complex interactions between atmospheric conditions, topography, and reflectance, significantly improving correction accuracy. Authors in [55, 85, 87] have also used AOD and CWV as additional inputs. Authors in [57] have used wind speed, surface barometric pressure, and different geometric angles to perform ocean regions' AC using Sentinel-3 OLCI.

- Radiative Transfer-Related Variables: Other studies have explored radiative transfer-related parameters as additional inputs to improve AC performance. Authors in [63] used total flux, diffuse transmittance, direct transmittance, spherical albedo, and path radiance, which are essential for modelling atmospheric radiative processes. Total flux represents the overall radiative energy received, influencing surface illumination conditions. Diffuse transmittance accounts for light scattered in all directions by atmospheric particles, while direct transmittance represents unscattered light travelling directly from the sun to the surface. Spherical albedo measures the reflectivity of the entire atmosphere, affecting the radiative balance and energy distribution. Path radiance represents the atmospheric scattering contributions to TOA radiance, which, if not corrected, can lead to an overestimation of SR. These parameters allow DL models to disentangle atmospheric effects from true SR, leading to a more physically consistent correction process across varying atmospheric conditions.

By integrating such physically meaningful variables, these models can disentangle atmospheric effects from true surface reflectance, resulting in more robust corrections across diverse environmental conditions.

4.2.2 Physics-Agnostic DL Models

Physics-agnostic AC models rely entirely on TOA reflectance without incorporating explicit physical constraints. These models are trained on large-scale paired TOA-BOA datasets, allowing them to extract complex correction patterns without requiring auxiliary atmospheric parameters.

Several DL-based physics-agnostic approaches have been developed for AC across remote sensing applications. Table 4 summarises different physics-aware DL approaches. Authors in [47] introduced a time-dependent encoder-decoder CNN for AC of vegetation reflectance in hyperspectral imaging (HSI). The model is trained on simulated datasets from SMARTS2 (for atmospheric and solar effects) and SCOPE (for vegetation reflectance). This method is beneficial for ground-based hyperspectral sensors capturing side-facing vegetation images, where atmospheric conditions are often unknown. Additionally, the date and time of image capture were included as inputs to account for solar position changes, atmospheric variations, and diurnal illumination effects, improving correction accuracy.

AC over coastal waters presents unique challenges due to the optical complexity of both aerosols and water, making traditional correction methods less effective. Authors in [65] tackled this issue by using simulated Rayleigh-corrected TOA data and training an ML model on datasets generated with AccuRT radiative transfer simulations. This approach accounts for many atmospheric and oceanic conditions, improving AC performance in coastal environments. The authors proposed a more advanced physics-agnostic model, DINSAT (Data-Driven Invertible Neural Surrogates of Atmospheric Transmission), in [66]. This method employs Neural Ordinary Differential Equations (Neural ODEs) to learn the underlying atmospheric physics and directly infer atmospheric transmission profiles from hyperspectral images. Unlike traditional AC models, DINSAT reconstructs transmission functions dynamically, making it adaptable to different atmospheric scenarios.

Other ML-based AC techniques include Gaussian Process AC (GPAC) and a Denoising Autoencoder (DA) introduced by the authors in [67]. These models were trained using MODTRAN-generated synthetic radiance-reflectance datasets, allowing them to learn ACs statistically without requiring explicit atmospheric parameters. GPAC leverages Bayesian regression to provide probabilistic corrections, while the DA treats atmospheric effects as noise and learns to remove them through an unsupervised learning framework.

Table 3 Summary of physics-aware DL models for AC (part-2)

Paper	Auxiliary Inputs Used (Other than TOA Radiance)	Physics Model	Dataset Used	Multispectral/ Hyperspectral	Spatial Process-ing (Pixel/ Image)	Target Environment	Model Used	End-to-End?	Result	Validation using Ground Data?
[64]	AOD, CWV, O3, NBAR, DEM, terrain characteristics (aspect, slope, shadows), PER1 sun–sensor acquisition geometry values	Sentinel 2	PERUSAT and Sentinel 2	Multispectral	Pixel	Land	Multiple Linear Regression and Feedforward network	Yes	RMSE: 0.012, r: 0.92	No
[85]	AOD, WVP	Sentinel 2	Sentinel 2	Multispectral	Image	Land	Diffusion model	Yes	RMSE: 0.0284, r: 0.95	No
[59]	SZA, VZA, RAA	SeaDAS software	MODIS Level-1 data and Software generated	Multispectral	Pixel	Ocean	Encoder-decoder with multihead attention	Yes	Test Results RMSE: 0.00021, r: 0.96; In-Situ Results RMSE: 0.0006, r: 0.30	MOBY measurements
[62]	SZA, SAA, VAA, AOD	6S	Py6S simulated	Any	Pixel	Land	MLP, 1-D CNN, ViT	No	RMSE: 0.018, r: 0.99	No
[86]	SZA, VZA, RAA	GOCI	GOCI, AERONET-OC for validation	Multispectral	Image	Ocean	Encoder-decoder with multihead attention	No	MAPD: 2.1, r: 0.97; In-Situ Results RMSE: 0.00016, r: 0.91; RMSE: 0.025	AERONET-OC
[87]	AOD, Water vapor, Flight path, imaging geometry	MODTRAN and LIBRADTRAN	AHIS	Hyperspectral	Pixel	Land	Random Forest	No	r: 0.99	No
[50]	SZA, VZA, RAA	GOCI	GOCI, AERONET-OC for validation, SeaDAS used for simulation	Multispectral	Pixel	Coastal Zone	Neural Network	Yes	MAPD: 16.76, r: 0.99; In-Situ Results MAPD: 27.87, r: 0.95	AERONET-OC
[56]	SAA, VZA, RAA, AOT	iCOR	Landsat 8, iCOR derived, in-situ	Multispectral	Pixel	Inland waters	CNN+Fully connected	Yes	RMSE: 0.005, MAPE: 1.5	On field Spectro-Radiometer
[57]	Wind speed, surface barometric pressure, SZA, angular info of observation geometry	MOMO	Sentinel −3 OLCI, in-situ collected using DALEC	Multispectral	Pixel	Coastal Zone	ANN	Yes	In-situ Results RMSE: 0.0013, r: 0.97	DALEC instrument

Low-altitude UAV remote sensing data is often affected by atmospheric scattering, haze, and absorption, degrading image quality and impacting reflectance retrieval. Traditional radiation correction methods struggle to recover true ground reflectance under such conditions [84]. Authors use a Transformer-based DL model- RCFormer leverages Vision Transformers (Swin Transformer) combined with parallel convolutional structures to enhance feature extraction and improve correction accuracy. In the context of hyperspectral AC, authors in [76] introduced a ML-based approach that combines Support Vector Machines (SVMs) for spectral feature learning with CNNs for spatial feature extraction using hypercolumn representations. For addressing seasonal variations in atmospheric effects, SAAC-Net was proposed as a Season-Aware DL Model for AC [30]. Trained on Landsat 8 imagery spanning different land covers, SAAC-Net incorporates seasonal information using a synthetic season band to model intra-annual changes in atmospheric properties. This season-aware correction strategy enhances model adaptability across different periods, making it more effective in correcting seasonal atmospheric distortions. Authors in [89] used images of the previous four years as an auxiliary input to make the model learn the temporal variations and spatial features. Authors in [77] have proposed onBOArd cloud detection and an AC network based on lightweight CNNs. Authors claimed their model performs 10 × faster than the operational S-2 L2A processor. The authors in [51] have proposed a DL model based on a neural network for AC in turbid waters. Turbid water is opaque due to suspended particles like clay or algae. The proposed NN model estimates SR for the NIR band for clear to turbid waters from the available SR values in the visible bands.

Despite their advantages, physics-agnostic AC models have limitations. Unlike physics-aware models, they do not explicitly incorporate prior knowledge about atmospheric interactions, making them less robust in extreme conditions. Additionally, their "black-box" nature raises concerns about interpretability, as they learn correction patterns implicitly rather than following well-defined physical laws. Nevertheless, their ability to process large-scale datasets efficiently, generalise across diverse atmospheric conditions, and eliminate the need for detailed atmospheric inputs makes them a promising alternative for real-time and scalable AC in remote sensing applications.

4.3 Design Principles for DL-Based Atmospheric Correction Models

Designing an effective DL-based AC (AC) model requires balancing physical interpretability, data-driven flexibility, and computational efficiency. The architecture must be aligned with the characteristics of the input data (e.g., spectral resolution, spatial coverage, temporal frequency) and the intended application domain (e.g., land, water, coastal, UAV).

Pixel-level networks such as 1D CNNs, FCNNs, and autoencoders (e.g., [55, 61, 63, 67]) are typically used when spectral fidelity is paramount and computational scalability is required for global AC. Image-level or spatial–spectral networks, including 2D/3D CNNs, R2U-Net, and Swin Transformer [49, 56, 76, 77, 82, 84], are preferred when spatial context and heterogeneous atmospheric effects must be captured—such as over mixed land–water scenes or urban–vegetation mosaics.

Integration of auxiliary physical parameters differentiates physics-aware from physics-agnostic designs. Physics-aware models (e.g., [49, 57, 61, 64]) demonstrate how geometric and atmospheric inputs—fused at early or intermediate layers—can constrain learning and improve generalizability across illumination and sensor configurations. In contrast, physics-agnostic models (e.g., [30, 47, 65, 66, 89]) achieve flexibility by learning directly from TOA–BOA pairs, often relying on extensive simulated or historical datasets to capture atmospheric variability.

Task-specific architectural adaptations are also common in the literature. For example, SAAC-Net [30] incorporates a synthetic "season band" to hanDLe intra-annual atmospheric variation, while DINSAT [66] leverages Neural ODEs to dynamically reconstruct transmission functions. These reflect a broader trend of embedding domain-specific priors—whether temporal, spectral, or physical—into the network design.

A critical consideration is generalizability. AC models trained on limited geographic or seasonal data risk overfitting to specific atmospheric conditions. Models trained on geographically or seasonally narrow datasets may fail in unseen conditions, as observed in UAV-based models like RCFormer [84] or coastal water corrections [51, 65] that perform well in niche domains but require retraining for other environments.

Uncertainty estimation is another emerging design priority. Bayesian DL, ensemble models, and dropout-based Monte Carlo inference can quantify confidence in corrections. While probabilistic approaches like GPAC [67] provide confidence estimates, most AC models remain deterministic, highlighting a gap in conveying correction reliability for operational decision-making.

A synthesis of these studies reveals several design principles in DL-based AC model development:

- Alignment of architecture with input characteristics: Model structures should be tailored to the data modality and scale, whether spectral-only or spatial–spectral,

Table 4 Summary of physics-agnostic DL models for AC

Paper	Auxiliary Inputs Used (Other than TOA Radiance)	Physics Model	Dataset Used	Multispectral/ Hyperspectral	Spatial Processing (Pixel/Image)	Target Environment	Model Used	End-to-End?	Result	Validation using Ground Data?
[47]	Date and Time	SCOPE and SMARTS2	Simulated Data using SCOPE - RTM is SAIL model	Hyperspectral	Pixel	Land	Time-dependent Encoder-Decoder CNN	Yes	RMSE: 0.134, r: 0.92	No
[65]	-	AccuRT	Generated using ACCuRT Model	Hyperspectral	Pixel	Coastal Zone	Neural Network, Partial least square regression	Yes	NRMSE: 0.045, r: 0.99	No
[82]	-	Sentinel-2	Sentinel-2	Multispectral	Image	Land	R2U-Net	Yes	RMSE: 0.0183, r: 0.94	No
[48]	Date and Time	MODTRAN	Modtran simulated data and Ground data	Multispectral	Pixel	Land	Time-dependent Encoder-Decoder CNN	No	Results are shown by graph	No
[66]	-	HyMap sensor	Target Detection Blind Test Project	Hyperspectral	Pixel	Land	MLP	Yes	PRMSE: 1.5%	No
[67]	-	MODTRAN	Self and simulated using MODTRAN	Hyperspectral	Pixel	Land	Gaussian process AC, Universal mean regression and Denoising Autoencoder	Yes	r: 0.96	No
[84]	-	NO	Self UAV captured	Multispectral	Image	Land	Swin Transformer MSAC and a feed-forward network	Yes	MSE: 0.0014, PSNR: 31.947	No
[76]	-	AVIRIS Sensor	AVIRIS Hyperspectral	Hyperspectral	Pixel	Land	CNN and SVM	No	RMSE: 0.03	No
[77]	-	LIBRADTRAN - Sen2Corr	Sentinel-2, AERONEL-Sen2 for validation	Multispectral	Image	Land	CNN	Yes	Testing Results: RMSE: 0.016; In-situ results: RMSE: 0.03	AERONET
[51]	-	MODISA	AERONET-OC for validation, SeaDAS used for simulation	Multispectral	Pixel	Turbid waters	Neural Network	Yes	Testing Results: RMSE: 0.0014, r: 0.97; In-situ results: RMSE: 0.0020, r: 0.77	AERONET-OC
[88]	-	LaSRC	Landsat 8	Multispectral	Image	Land	Encoder-Decoder	Yes	RMSE: 0.0282, r: 0.94	No
[30]	-	LaSRC	Landsat 8	Multispectral	Image	Land	CNN with Residual Blocks	Yes	RMSE: 0.0042, r: 0.99; In-situ results: RMSE: 0.053, r: 0.88	RadCalNet

Table 4 (continued)

Paper	Auxiliary Inputs Used (Other than TOA Radiance)	Physics Model	Dataset Used	Multispectral/ Hyperspectral	Spatial Processing (Pixel/Image)	Target Environment	Model Used	End-to-End?	Result	Validation using Ground Data?
[89]	Optical flow of Previous year images	LaSRC	Landsat 8	Multispectral	Image	Land	CNN with spatial and temporal channel	Yes	Testing Results: RMSE: 0.0017, r: 0.88; In-situ results: MRD: 0.027	RadCalNet

hyperspectral or multispectral, and pixel-level or patch-based, to ensure optimal exploitation of available information.

- Integration of domain priors: Incorporating prior knowledge—through explicit physical parameters in physics-aware models or through temporal and seasonal features in physics-agnostic models—enhances model robustness under variable atmospheric conditions.
- Incorporation of task-specific enhancements: Architectural adaptations such as synthetic season bands, hypercolumn feature representations, or Neural Ordinary Differential Equation (ODE) layers can address unique atmospheric or environmental challenges, improving correction accuracy.
- Planning for generalisation: Training strategies should include multi-region and multi-season datasets and augmentation techniques that replicate diverse atmospheric states to improve performance across unseen conditions.
- Quantification of predictive uncertainty: Emerging approaches, such as Gaussian Process AC and ensemble-based inference, offer probabilistic outputs that can improve operational reliability and foster user trust in DL-based AC products.

4.4 Assumptions in DL-Based AC Methods

DL-based AC models rely on certain assumptions to simplify complex physical processes and reduce computational overhead. While these assumptions improve computational efficiency and allow for large-scale deployment, they also introduce potential limitations that may affect model generalisation under complex atmospheric conditions.

In the context of DL-based AC, we identify four key categories of assumptions:

1. Assumptions-1: Simplifying RTE
2. Assumptions-2: Fixed atmospheric conditions
3. Assumptions-3: Clear-sky condition
4. Assumptions-4: Lambertian surface.

4.4.1 Assumptions-1: Simplifying RTE

The first category of assumptions involves simplifications in the RTE, where DL models avoid solving the full radiative transfer problem by approximating it with empirical or data-driven formulations. Many DL-based AC models assume a linear or near-linear relationship between TOA radiance and bottom-of-atmosphere reflectance [55]. This simplification is based on the idea that atmospheric effects can be learned as a function of TOA reflectance and auxiliary atmospheric parameters rather than explicitly solving the RTE for each observation. Additionally, specific models assume uniform aerosol scattering and single-layer atmospheric absorption, neglecting multiple scattering effects, leading to inaccuracies in regions with high aerosol optical thickness or complex atmospheric compositions. These models achieve real-time AC performance using data-driven parameterisation rather than full radiative transfer computations. Still, they may suffer from reduced accuracy in highly dynamic atmospheric conditions.

4.4.2 Assumptions-2: Fixed Atmospheric Conditions

The second major assumption category pertains to atmospheric conditions. This assumption is especially prevalent in studies that rely on synthetic datasets generated using radiative transfer models such as MODTRAN, 6S, or AccuRT. In such simulations, users must define fixed atmospheric profiles—commonly mid-latitude summer, tropical, or U.S. Standard Atmosphere—which prescribe static values for aerosol optical properties, ozone, carbon dioxide, and water vapour content. These predefined profiles are then applied uniformly across all simulated scenes, allowing for tractable training data generation but at the cost of atmospheric realism. Consequently, models trained under these assumptions may not generalise well to real-world scenarios where atmospheric composition and structure vary spatially and temporally.

4.4.3 Assumptions-3: Clear-Sky Condition

The clear-sky condition is often implicitly assumed unless explicitly hanDLed in the model pipeline. Many studies exclude cloud-affected pixels or do not incorporate any cloud masking or detection module, thereby training and evaluating models only on cloud-free imagery. While this reduces the complexity of modelling atmospheric interactions, it also limits the applicability of such models in regions prone to persistent cloud cover, such as tropical and coastal zones, where cloud-induced scattering and absorption substantially affect SR retrievals.

Table 5 Assumptions followed by physics-aware DL-based AC models

Paper	Assumption-1	Assumption-2	Assumption-3	Assumption-4
[61]	✗	✓	✓	✗
[75]	✗	✗	✓	✓
[78]	✗	✗	✗	✓
[58]	✗	✓	✓	✓
[63]	✗	✗	✓	✓
[68]	✓	✓	✓	✓
[49]	✓	✓	✓	✓
[55]	✓	✗	✓	✓
[64]	✗	✗	✓	✓
[85]	✗	✗	✓	✓
[59]	✓	✓	✓	✓
[62]	✗	✓	✓	✓
[86]	✓	✓	✗	✓
[87]	✓	✓	✓	✓
[50]	✓	✓	✓	✓
[56]	✗	✗	✓	✓
[57]	✓	✓	✓	✓

4.4.4 Assumptions-4: Lambertian Surface

Additionally, bidirectional reflectance distribution function (BRDF) effects are often ignored in AC models, as most assume Lambertian surfaces, wherein the surface reflects incident radiation uniformly in all directions, independent of the illumination and viewing geometry. This simplification is commonly employed to reduce model complexity by ignoring BRDF effects, which describe the angular dependence of SR. While the Lambertian assumption may be a reasonable approximation for certain surfaces under specific conditions, it does not hold in many real-world scenarios. Most natural surfaces—including vegetation, snow, bare soil, and water—exhibit non-Lambertian or anisotropic reflectance behaviour. Only a few studies, such as [61], incorporate BRDF effects or correct for anisotropic reflectance. Here, we have considered the Lambertian surface assumption to be adopted in all studies that do not explicitly address BRDF corrections. It is to be noted that while the inclusion of SZA, VZA, and RAA in DL-based AC models provides the geometric context to the model, it does not necessarily imply that BRDF effects are explicitly hanDLed unless the model is specifically designed or validated to account for directional reflectance behaviour.

Tables 5 and 6 provide an overview of the assumptions followed by the surveyed articles. A checkmark (✓) indicates the presence of the assumption, while a cross (×) denotes its absence.

5 Open Issues and Future Directions

Despite significant advancements in DL-based AC, several challenges hinder its widespread adoption and operational deployment. These challenges primarily relate to model generalisation, computational efficiency, dataset availability, cross-sensor applicability, and benchmarking. Addressing these issues is essential for improving the scalability, robustness, and interpretability of DL models for AC across diverse environmental conditions.

Table 6 Assumptions followed by physics agnostic DL-based AC models

Paper	Assumption-1	Assumption-2	Assumption-3	Assumption-4
[47]	✗	✓	✓	✓
[65]	✗	✓	✓	✓
[82]	✗	✗	✓	✓
[48]	✓	✗	✓	✓
[66]	✓	✗	✓	✓
[67]	✗	✓	✓	✓
[84]	✗	✗	✓	✓
[76]	✓	✗	✓	✓
[77]	✗	✓	✗	✓
[51]	✓	✓	✓	✓
[88]	✗	✗	✓	✓
[30]	✗	✗	✓	✓
[89]	✗	✗	✓	✓

5.1 Model Generalisation Across Atmospheric and Geographic Variability

A persistent challenge in DL-based AC is achieving robust generalisation across diverse atmospheric conditions, geographic regions, land cover types, and sensor platforms. Models trained on geographically, seasonally, or sensor-specific datasets may degrade when deployed in different environments or under previously unseen atmospheric conditions. Future research should prioritise cross-domain generalisation by incorporating diverse, multi-region, and multi-season datasets, domain adaptation and transfer learning techniques.

Atmospheric composition differs significantly across regions, altitudes, and proximity to pollution sources. This leads to performance degradation when models are applied beyond their training domain. For instance, the SAAC-Net model [30], trained on diverse Indian land covers, achieved an RMSE of 0.043 in native conditions but showed higher errors when tested in Dubai (RMSE = 0.067) and at a Railroad site in the USA (RMSE = 0.096). Achieving global generalisation requires training on datasets that capture a broad spectrum of geographic and seasonal variability. However, collecting concurrent TOA–BOA observations across such diversity is logistically challenging. To address this, transfer learning approaches and generative models—such as diffusion-based data synthesis—could be leveraged to expand training datasets with geographically diverse yet physically consistent samples.

Variability in land cover types presents another dimension of complexity. Distinct reflectance properties across vegetation, urban, desert, and aquatic surfaces influence how atmospheric distortions are modelled. A model optimised for vegetation-dominated regions may underperform in spectrally unique areas such as urban valleys or plantation landscapes. For example, SAAC-Net [30] exhibited an RMSE of 0.096 at the Railroad site in the USA and 0.051 in Kerala's plantation-covered areas during winter—both higher than its baseline performance—indicating sensitivity to surface type and seasonal dependencies. Future work should prioritise training datasets with balanced representation of diverse land covers and seasons. Multi-task learning frameworks, where AC models jointly learn auxiliary tasks such as land cover classification, could further improve adaptability across heterogeneous surfaces.

A final constraint on generalisation arises from sensor-specific characteristics. Each satellite sensor has distinct spectral response functions and radiometric calibration, making cross-sensor transfer non-trivial. Models trained exclusively on one sensor often fail to generalise to others without significant retraining. Research should focus on spectral harmonisation techniques and cross-sensor transfer learning frameworks to address this. Such methods could enable sensor-agnostic AC models capable of producing consistent surface reflectance across multiple platforms, thereby supporting long-term, multi-sensor environmental monitoring.

5.2 Computational Efficiency and Real-Time Processing

Computational efficiency remains a critical bottleneck for large-scale AC. Satellite imagery is acquired at high spatial and temporal resolutions, necessitating near real-time processing capabilities for environmental monitoring and disaster response applications. However, DL-based AC models, particularly those with complex architectures, are computationally intensive, requiring significant hardware resources for training and inference.

Model compression techniques such as pruning, quantisation, and knowledge distillation should be explored to improve computational efficiency. Additionally, lightweight architectures optimised for edge computing can facilitate deployment on resource-constrained platforms, such as drones and onBOArd satellite processors. Authors in [30] deployed the SAAC-Net AC model on the NVIDIA Jetson Orin platform using TensorRT-based compression and optimisation. While inference time was significantly reduced, the RMSE increased nearly fivefold, demonstrating a tradeoff between computational efficiency and model performance. Future research should focus on developing hardware-aware neural architectures that balance accuracy and efficiency for operational deployment.

5.3 Uncertainty Quantification for Decision-Making

Future DL-based AC systems should move beyond point predictions to provide probabilistic outputs, enabling downstream applications (e.g., change detection, vegetation monitoring) to factor correction confidence into their decision pipelines. Incorporating Bayesian inference, ensemble modelling, and Monte Carlo dropout can provide actionable uncertainty estimates while preserving computational tractability.

5.4 Addressing Dataset Shift in AC

Dataset shift is another major challenge that impacts model reliability during deployment. The data encountered in real-world operational use may differ from training data due to evolving atmospheric conditions, seasonal variations, and land cover changes. This discrepancy leads to performance degradation, limiting the effectiveness of DL models in dynamic environments. To quantify dataset shift, authors in [96] conducted statistical analyses demonstrating that land cover-specific RMSE values increase by 10% due to variations in atmospheric properties over time. To address this, spatiotemporal DL models have been proposed [89], incorporating historical images (e.g., four years of prior observations) to model interannual atmospheric changes. However, a four-year period may be insufficient capture long-term atmospheric trends fully. A more effective approach would involve continual learning frameworks, where models are periodically updated with new observations to adapt to changing atmospheric conditions. Active learning strategies, where models selectively query new training samples from operational data, could mitigate dataset shifts and enhance long-term performance.

5.5 Lack of Benchmark Datasets and Standardised Evaluation Metrics

One of the most pressing issues in DL-based AC is the absence of standardised benchmarking datasets and evaluation protocols. Many studies use proprietary datasets, making objectively comparing different approaches difficult. Additionally, no universally accepted set of evaluation metrics leads to inconsistencies in model performance assessments. Establishing benchmark datasets encompassing diverse atmospheric conditions, geographic regions, and land cover types is essential for fostering reproducibility in DL-based AC research. Standardising evaluation metrics, such as Root Mean Square Error (RMSE) and Spectral Angle Mapping (SAM), will facilitate fair and meaningful comparisons between different models.

5.6 Limitations of Physics-Based Training Data

Most DL models for AC are trained using TOA-BOA paired datasets generated by physics-based AC models. For example, models in [30], [82], and [78] are trained using reflectance data from LASRC, Sen2Cor, and MAIAC. This reliance inherently limits model accuracy to that of the physics-based models themselves. Ideally, ground-truth SR data should be obtained using on-site spectrometers or drones to provide high-fidelity training datasets. However, large-scale ground-truth data collection is logistically and financially challenging, requiring simultaneous TOA-BOA observations across diverse land covers and atmospheric conditions. To mitigate this issue, stable diffusion techniques can generate synthetic datasets that better approximate real-world conditions, effectively supplementing limited ground-based observations.

6 Conclusion

This paper presented the first comprehensive systematic review of DL-based AC methods for satellite remote sensing. It outlined the evolution from traditional image-based and physics-based approaches to data-driven DL models and categorised current methods into physics-aware and physics-agnostic frameworks. The review investigated the wide range of datasets used for training and validation—including satellite data, simulation-generated data, and field campaigns—and provided a comparative analysis of DL architectures and their input configurations, such as atmospheric parameters, solar-sensor geometry, and TOA reflectance.

Key contributions of this review include a structured classification of DL approaches, detailed summaries of 30 papers with unified evaluation criteria, and identification of core assumptions (e.g., fixed atmospheric profiles, clear-sky conditions, Lambertian surfaces). Despite promising results, current DL-based AC models face limitations in generalisability, especially across geographic regions, atmospheric states, land cover types, and sensors. Furthermore, challenges such as dataset shift, lack of benchmark datasets, and limited temporal adaptability hinder large-scale deployment. To overcome these, the paper recommends future research into hybrid physics-DL models, cross-domain transfer learning, synthetic data generation, continual learning, and standardisation of datasets and evaluation protocols.

By integrating and synthesising dispersed research efforts, this review provides a comprehensive roadmap for advancing DL approaches in AC, ultimately supporting more reliable and scalable Earth observation analytics.

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Declarations

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References

1. Nijhawan R, Joshi D, Narang N, Mittal A, Mittal A (2019) A futuristic deep learning framework approach for land use-land cover classification using remote sensing imagery. In: Mandal J, Bhattacharyya D, Auluck N (eds) Advanced computing and communication technologies. Springer Singapore, Singapore, pp 87–96
2. Landsat missions.: example of landsat collection 1 surface reflectance. Online: https://www.usgs.gov/media/images/example-landsat-collection-1-surface-reflectance
3. Potapov P, Hansen MC, Kommareddy I, Kommareddy A, Turubanova S, Pickens A et al. (2020) Landsat analysis ready data for global land cover and land cover change mapping. Remote Sens 12(3). https://doi.org/10.3390/rs12030426
4. Dwyer JL, Roy DP, Sauer B, Jenkerson CB, Zhang HK, Lymburner L (2018) Analysis ready data: enabling analysis of the landsat archive. Remote Sens 10(9). https://doi.org/10.3390/rs10091363
5. Giuliani G, Chatenoux B, Honeck E, Richard JP (2018) Towards sentinel-2 analysis ready data: a Swiss data cube perspective. In IGARSS 2018 – 2018 IEEE International Geoscience and Remote Sensing Symposium, pp 8659–8662
6. CEOS.: CEOS analysis ready data. Accessed: 2024-04-3. https://ceos.org/ard/
7. Committee on Earth Observation Satellites (CEOS).: CARD4L product family specification: surface reflectance. Version 5.0. https://ceos.org/ard/files/Self%20Assessments/SR/v5.0/CARD4L_Product_Family_Specification_Surface_Reflectance-v5.0_S2_SelfAssessment_20211124.pdf
8. Doxani G, Vermote EF, Roger JC, Skakun S, Gascon F, Collison A et al. (2023) Atmospheric correction inter-comparison eXercise, ACIX-II land: an assessment of atmospheric correction processors for landsat 8 and sentinel-2 over land. Remote Sens of Environ 285:113412. https://doi.org/10.1016/j.rse.2022.113412
9. Kganyago M, Mhangara P, Adjorlolo C (2021) Estimating crop biophysical parameters using machine learning algorithms and sentinel-2 imagery. Remote Sens 13(21). https://doi.org/10.3390/rs13214314
10. Parida BR, Kumari A (2021) Mapping and modeling mangrove biophysical and biochemical parameters using sentinel-2A satellite data in Bhitarkanika National Park, Odisha. Model Earth Syst Environ 7:2463–2474. https://doi.org/10.1007/s40808-020-01005-3
11. Patel U, Pathan M, Kathiria P, Patel V (2023) Crop type classification with hyperspectral images using deep learning: a transfer learning approach. Model Earth Syst Environ 9:1977–1987. https://doi.org/10.1007/s40808-022-01608-y
12. Brando VE, Dekker AG (2003) Satellite hyperspectral remote sensing for estimating estuarine and coastal water quality. IEEE T Geosci Remote 41(6):1378–1387. https://doi.org/10.1109/TGRS.2003.812907
13. Ramaraj M, Sivakumar R (2023) Integration of band regression empirical water quality (BREWQ) model with deep learning algorithm in spatiotemporal modeling and prediction of surface water quality parameters. Model Earth Syst Environ 9:3279–3304. https://doi.org/10.1007/s40808-023-01695-5
14. Saeidi V, Seydi ST, Kalantar B, Ueda N, Tajfirooz B, Shabani F (2023) Water depth estimation from sentinel-2 imagery using advanced machine learning methods and explainable artificial intelligence. Geomatics, Nat Hazards and Risk 14(1):2225691. https://doi.org/10.1080/19475705.2023.2225691
15. Angelopoulou T, Tziolas N, Balafoutis A, Zalidis G, Bochtis D (2019) Remote sensing techniques for soil organic carbon estimation: a review. Remote Sens 11(6). https://doi.org/10.3390/rs11060676
16. Verma P, Raghubanshi AS (2019) Rural development and land use land cover change in a rapiDLy developing agrarian South Asian landscape. Remote Sens Appl: Soc and Environ. 14:138–147. https://doi.org/10.1016/j.rsase.2019.03.002
17. Sabri A, Bahi H, Bounoua L, Tahiri M, Tweed S, LeBlanc M et al. (2024) Predictive modelling on spatial–temporal land use and land cover changes at the casablanca-settat region in Morocco. Model Earth Syst Environ 10:6691–6714. https://doi.org/10.1007/s40808-024-02107-y
18. Chavez PS (1988) An improved dark-object subtraction technique for atmospheric scattering correction of multispectral data. Remote Sens of Environ 24(3):459–479. https://doi.org/10.1016/0034-4257(88)90019-3
19. Smith GM, Milton EJ (1999) The use of the empirical line method to calibrate remotely sensed data to reflectance. Int J Remote Sens 13:2653–2662
20. Vermote EF, Tanre D, Deuze JL, Herman M, Morcette JJ (1997) Second simulation of the satellite signal in the solar spectrum, 6S: an overview. IEEE T Geosci Remote 35(3):675–686. https://doi.org/10.1109/36.581987
21. Berk A, Anderson GP, Bernstein LS, Acharya PK, Dothe H, Matthew MW et al. (1999) MODTRAN4 radiative transfer modeling for atmospheric correction. In: Optical spectroscopic techniques and instrumentation for atmospheric and space research iii., vol 3756. SPIE, pp 348–353
22. Richter R, Schläpfer D (2019) Atmospheric and topographic correction (ATCOR theoretical background document). DLR Ib 1:0564–03
23. Zhong L, Lei H, Yang J (2024) Development of a distributed physics-informed deep learning hydrological model for data-scarce regions. Water Resour Res 60. https://doi.org/10.1029/2023WR036333
24. Han W, He TL, Jiang Z, Zhu R, Jones D, Miyazaki K et al. (2023) The capability of deep learning model to predict ozone across continents in China, the United States and Europe. Geophys Res Lett 50. https://doi.org/10.1029/2023GL104928
25. Rasp S, Pritchard M, Gentine P (2018) Deep learning to represent subgrid processes in climate models. Proc Natl Acad Sci USA 115:9684–9689. https://doi.org/10.1073/pnas.1810286115
26. Nguyen T, Brandstetter J, Kapoor A, Gupta JK, Grover A ClimaX: a foundation model for weather and climate. Available from: https://arxiv.org/abs/2301.10343

27. Hernández D, Cecilia JM, Cano JC, Calafate C (2022) Flood detection using real-time image segmentation from unmanned aerial vehicles on edge-computing platform. Remote Sens 14:223. https://doi.org/10.3390/rs14010223
28. Liu J, Xiang J, Jin Y, Liu R, Yan J, Wang L (2021) Boost precision agriculture with unmanned aerial vehicle remote sensing and edge intelligence: a survey. Remote Sens 13:4387. https://doi.org/10.3390/rs13214387
29. Koubâa A, Ammar A, Abdelkader M, AlHabashi Y, Ghouti L (2023) Aero: AI-Enabled remote sensing observation with onBOArd edge computing in UAVs. Remote Sens 15:1873. https://doi.org/10.3390/rs15071873
30. Shah M, Raval MS, Divakaran S, Dhar D, Parmar H (2023) SAAC-Net: deep neural network-based model for atmospheric correction in remote sensing. Int J Remote Sens 44(23):7365–7389. https://doi.org/10.1080/01431161.2023.2282407
31. Vermote E, Justice C, Claverie M, Franch B (2016) Preliminary analysis of the performance of the landsat 8/OLI land surface reflectance product. Remote Sens of Environ 185:46–56. Landsat 8 Science Results. doi: https://doi.org/10.1016/j.rse.2016.04.008
32. Moher D, Liberati A, Tetzlaff J, Altman DG, Group P et al. (2010) Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Int J Surg 8(5):336–341
33. Stamnes K, Hamre B, Stamnes S, Chen N, Fan Y, Li W et al. (2018) Progress in forward-inverse modeling based on radiative transfer tools for coupled atmosphere-snow/Ice-ocean systems: a review and description of the AccuRT model. Appl Sci 8(12). https://doi.org/10.3390/app8122682
34. Yang P, Prikaziuk E, Verhoef W, van der Tol C (2021) Scope 2.0: a model to simulate vegetated land surface fluxes and satellite signals. Geoscientific Model Devel 14(7):4697–4712. https://doi.org/10.5194/gmd-14-4697-2021
35. European Space Agency (ESA).: sentinel-2 user handbook. https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi. Online
36. Justice CO et al. (1998) The moderate resolution imaging spectroradiometer (modis): land remote sensing for global change research. IEEE T Geosci Remote 36(4):1228–1249
37. Roy DP, Wulder MA, Loveland TR, W CE, Allen RG, Anderson MC et al. (2014) Landsat-8: science and product vision for terrestrial global change research. Remote Sens of Environ 145. https://doi.org/10.1016/j.rse.2014.02.001
38. Green RO, Eastwood ML, Sarture CM, Chrien TG, Aronsson M, Chippendale BJ et al. (1998) Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS). Remote Sens of Environ 65(3):227–248. https://doi.org/10.1016/S0034-4257(98)00064-9
39. Pearlman JS, Carman SL, Segal CC et al. (2003) Hyperion, a space-based imaging spectrometer. IEEE T Geosci Remote 41(6):1160–1173
40. European Space Agency (ESA).: sentinel-3 user guide. https://sentiwiki.copernicus.eu/web/olci-processing. Online
41. Spectral imaging Ltd.: hyperspectral camera aisa Eagle (specim). Online: https://spaces.awi.de/spaces/RA/pages/318727494/Hyperspectral+Camera+AISA+Eagle+Specim
42. Bessho K, Date K, Hayashi M, Ikeda A, Imai T, Inou H et al. (2016) An introduction to himawari-8/9 - Japan's new-generation geostationary meteorological satellites. J Meteorolog Soc Jpn Ser II. 94(2):151–183. https://doi.org/10.2151/jmsj.2016-009
43. Choi JK, Park YJ, Ahn JH, Lim HS, Eom J, Ryu JH (2012) GOCI, the world's first geostationary ocean color observation satellite, for the monitoring of temporal variability in coastal water turbidity. J Educ Chang Geophys Res: Oceans 117(C9). https://doi.org/10.1029/2012JC008046
44. eoPortal directory.: PeruSAT-1 earth observation minisatellite. Satellite missions catalogue. https://www.eoportal.org/satellite-missions/perusat-1#naomi-new-astrosat-optical-modular-instrument. Online
45. Liu J, Ye X, Song Q, Ding J, Zou B (2023) Products of HY-1C/D ocean color satellites and their typical applications. Natl Remote Sens Bull. 27(1):1–13. https://doi.org/10.11834/jrs.20235002
46. Indian Space Research Organization (ISRO).: isro Resourcesat 1 and Resourcesat 2 - LISS-3. https://www.data.gov.in/catalog/liss-iii-data-download-indian-remote-sensing-satellite-resourcesat-1. Online
47. Qamar F, Dobler G (2023) Atmospheric correction of vegetation reflectance with simulation-trained deep learning for ground-based hyperspectral remote sensing. Plant Methods 12; 19. https://doi.org/10.1186/s13007-023-01046-6
48. Sun J, Xu F, Cervone G, Gervais M, Wauthier C, Salvador M (2021) Automatic atmospheric correction for shortwave hyperspectral remote sensing data using a time-dependent deep neural network. ISPRS J Photogramm 174:117–131. https://doi.org/10.1016/j.isprsjprs.2021.02.007
49. Xu F, Cervone G, Franch G, Salvador M (2020) Multiple geometry atmospheric correction for image spectroscopy using deep learning. J Appl Remote Sens 14:1. https://doi.org/10.1117/1.jrs.14.024518
50. Men J, Yao T, Yang C, Tian L (2024) A neural network-based atmospheric correction algorithm for GOCI imagery over coastal waters. IEEE T Geosci Remote 62:1–16. https://doi.org/10.1109/TGRS.2023.3339619
51. Wang J, Wang Y, Lee Z, Wang D, Chen S, Lai W (2022) A revision of nasa SeaDAS atmospheric correction
52. Bouvet M, Thome KJ, Berthelot B, Bialek A, Czapla-Myers JS, Fox NP et al. (2019) RadCalNet: a radiometric calibration network for Earth observing imagers operating in the visible to shortwave infrared spectral range. Remote Sens 11:2401
53. Zibordi G, Mélin F, Berthon JF, Holben B, Slutsker I, Giles D et al. (2009) AERONET-OC: a network for the validation of ocean color primary products. J Atmos Ocean Technol 26(8):1634–1651. https://doi.org/10.1175/2009JTECHO654.1
54. .: AERONET-OC. https://aeronet.gsfc.nasa.gov/new_web/ocean_color.html. OnlineNASA
55. Rusia P, Bhateja Y, Misra I, Moorthi SM, Dhar D (2021 10) An efficient machine learning approach for atmospheric correction. J Indian Soc Remote Sens 49:2539–2548. https://doi.org/10.1007/s12524-021-01406-6
56. Nguyen MV, La OT, Nguyen HTT, Heriza D, Lin BY, Ryadi GYI et al. (2025) Landsat 8 oli atmospheric correction neural network for inland waters in tropical regions. Int J Environ Sci Technol 22:6769–6788. https://doi.org/10.1007/s13762-024-06080-y
57. Schroeder T, Schaale M, Lovell J, Blondeau-Patissier D (2022) An ensemble neural network atmospheric correction for sentinel-3 olci over coastal waters providing inherent model uncertainty estimation and sensor noise propagation. Remote Sens of Environ 270:112848. https://doi.org/10.1016/j.rse.2021.112848
58. Men J, Liu J, Xia G, Yue T, Tong R, Tian L et al. (2022) Atmospheric correction for HY-1C czi images using neural network in western Pacific region. Geo Spat Inf Sci 25:476–488. https://doi.org/10.1080/10095020.2021.2009314
59. Men J, Tian L, Zhao D, Wei J, Feng L (2022) Development of a deep learning-based atmospheric correction algorithm for oligotrophic oceans. IEEE T Geosci Remote 60:1–19. https://doi.org/10.1109/TGRS.2022.3215767
60. Jain A Understanding the 1D convolutional layer in deep learning. Accessed: 2025-04-15. https://medium.com/@abhishekjainindore24/understanding-the-1d-convolutional-layer-in-deep-learning-7a4cb994c981
61. Ojaghi S, Bouroubi Y, Foucher S, Bergeron M, Seynat C (2023) Deep learning-based emulation of radiative transfer models

for top-of-atmosphere BRDF modelling using sentinel-3 olci. Remote Sens 15(3). https://doi.org/10.3390/rs15030835
62. Jasso-Garduño AE, Muñoz-Máximo I, Pinto D, Ramírez-Cortés JM (2024) Deep learning based emulation of radiative transfer code for atmospheric correction of satellite images. Computacion y Sistemas 28:2327–2341. https://doi.org/10.13053/CyS-28-4-5287
63. Pyo JC, Duan H, Ligaray M, Kim M, Baek S, Kwon YS et al. (2020) An integrative remote sensing application of stacked autoencoder for atmospheric correction and cyanobacteria estimation using hyperspectral imagery. Remote Sens. 4;12. https://doi.org/10.3390/rs12071073
64. Saldarriaga L, Tan Y, Sabater N, Delegido J (2025) A pixel-based machine learning atmospheric correction for PeruSAT-1 imagery. Remote Sens 2:17. https://doi.org/10.3390/rs17030460
65. Borge OM, Bakken S, Johansen TA (2021) Atmospheric correction of hyperspectral data over coastal waters based on machine learning models. In Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing. vol. 2021-March, IEEE Computer Society
66. Koch J, Forland B, Bernacki B, Doster T, Emerson T (2024) Data-driven invertible neural surrogates of atmospheric transmission. In IGARSS 2024 – 2024 IEEE International Geoscience and Remote Sensing Symposium, pp 6943–6947
67. Basener B, Basener A (2023) Gaussian process and deep learning atmospheric correction. Remote Sens 15(3). https://doi.org/10.3390/rs15030649
68. Fan Y, Li W, Gatebe CK, Jamet C, Zibordi G, Schroeder T et al. (2017 9) Atmospheric correction over coastal waters using multilayer neural networks. Remote Sens of Environ 199:218–240. https://doi.org/10.1016/j.rse.2017.07.016
69. Krizhevsky A, Sutskever I, Hinton GE (2012) ImageNet classification with deep convolutional neural networks. Adv Neural Inf Process Syst 1097–1105
70. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. CoRR. (https://arxiv.org/abs/1409.1556)
71. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 770–778
72. Kattenborn T, Eichel J, Fassnacht FE (2019) Convolutional neural networks enable efficient, accurate and fine-grained segmentation of plant species and communities from high-resolution uav imagery. Sci Rep 9:17656. https://doi.org/10.1038/s41598-019-53797-9
73. Song J, Gao S, Zhu Y, Ma C (2019) A survey of remote sensing image classification based on CNNs. Big Earth Data 3(3):232–254. https://doi.org/10.1080/20964471.2019.1657720
74. Hoeser T, Bachofer F, Kuenzer C (2020) Object detection and image segmentation with deep learning on earth observation data: a review—part ii: applications. Remote Sens 12(18). https://doi.org/10.3390/rs12183053
75. Zhao X, Ma Y, Xiao Y, Liu J, Ding J, Ye X et al. (2023 11) Atmospheric correction algorithm based on deep learning with spatial-spectral feature constraints for broadband optical satellites: examples from the HY-1C coastal zone imager. ISPRS J Photogramm 205:147–162. https://doi.org/10.1016/j.isprsjprs.2023.10.006
76. Zhu S, Lei B, Wu Y (2018 2 10) Retrieval of hyperspectral surface reflectance based on machine learning. Remote Sens. https://doi.org/10.3390/rs10020323
77. Mateo-García G, Aybar C, Acciarini G, Růžička V, Meoni G, Longépé N et al. (2023) OnBOArd cloud detection and atmospheric correction with deep learning emulators. In IGARSS 2023 – 2023 IEEE International Geoscience and Remote Sensing Symposium, pp 1875–1878
78. Duffy K, Vandal TJ, Wang W, Nemani RR, Ganguly AR (2023 7) A framework for deep learning emulation of numerical models with a case study in satellite remote sensing. IEEE Trans on Neural Networks and Learn Syst 34:3345–3356. https://doi.org/10.1109/TNNLS.2022.3169958
79. Ronneberger O, Fischer P, Brox T (2015) U-net: convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer-assisted intervention, Springer. p. 234–241
80. Solórzano JV, Mas JF, Gao Y, Gallardo-Cruz JA (2021) Land use land cover classification with U-Net: advantages of combining sentinel-1 and sentinel-2 imagery. Remote Sens 13(18). https://doi.org/10.3390/rs13183600
81. Badrinarayanan V, Kendall A, Cipolla R (2017) SegNet: a deep convolutional encoder-decoder architecture for image segmentation. IEEE T Pattern Anal 39(12):2481–2495. https://doi.org/10.1109/TPAMI.2016.2644615
82. Langstad K, Völgyes D, Parelius EJ (2024) Atmospheric corrections for earth observation with deep learning. Institute of Electrical and Electronics Engineers (IEEE), pp 7383–7387
83. Liu Z, Lin Y, Cao Y, Hu H, Wei Y, Zhang Z et al. (2021) Swin transformer: hierarchical vision transformer using shifted windows. In 2021 IEEE/CVF International Conference on Computer Vision (ICCV), pp 9992–10002
84. Liu T, Zhang Y, QH (2025) Rcformer: radiation correction method for degraded multispectral uav images using vision transformers. Int J Remote Sens 46(5):2192–2211. https://doi.org/10.1080/01431161.2024.2449476
85. Li A, Guan G, Zhao H, Li S, Zhu J, Han X et al. (2024) Integrated methodology for atmospheric correction and cloud removal of multispectral remote sensing images using guided diffusion model. IEEE T Geosci Remote 62:1–21. https://doi.org/10.1109/TGRS.2024.3497180
86. Men J, Feng L, Chen X, Tian L (2023) Atmospheric correction under cloud edge effects for geostationary ocean color imager through deep learning
87. Zhou Q, Wang S, Liu N, Townsend PA, Jiang C, Peng B et al. (2022) Towards operational atmospheric correction of airborne hyperspectral imaging spectroscopy: algorithm evaluation, key parameter analysis, and machine learning emulators
88. Shah M, Raval MS, Divakaran S (2021) Deep learning-based emulator for 6S atmospheric correction model. In 2021 IEEE International India Geoscience and Remote Sensing Symposium (InGARSS), pp 471–474
89. Shah M, Raval MS, Divakaran S, Dhar D, Parmar H (2025) A spatio-temporal deep learning model for enhanced atmospheric correction. Model Earth Syst Environ 11. https://doi.org/10.1007/s40808-024-02175-0
90. Brajard J, Santer R, Crépon M, Thiria S (2012) Atmospheric correction of meris data for case-2 waters using a neuro-variational inversion. Remote Sens of Environ 11;126:51–61. https://doi.org/10.1016/j.rse.2012.07.004
91. Tan W, He H, Chen X, Qi W (2020) Analyzing the influence of atmosphere on optical remote sensing in 400 to 2500 nm wavelength spectrum. In: Liu J, Wang Y, Ling Z, Sun Y, Jin D (eds) AOPC 2020: optical spectroscopy and Imaging; and biomedical optics., vol 11566. International Society for Optics and Photonics. SPIE, p 115660I. Available from: https://doi.org/10.1117/12.2579576
92. Engelen R, Stephens G (1999) Characterization of water-vapour retrievals from TOVS/HIRS and SSM/T-2 measurements. Quart J Roy Meteor Soc 125:331–351. https://doi.org/10.1002/QJ.49712555317
93. Preusker R, Henken CC, Fischer J (2021) Retrieval of daytime total column water vapour from olci measurements over land surfaces. Remote Sens 13:932. https://doi.org/10.3390/rs13050932

94. Richter R, Schläpfer D (2008) Considerations on water vapor and surface reflectance retrievals for a spaceborne imaging spectrometer. IEEE T Geosci Remote 46:1958–1966. https://doi.org/10.1109/TGRS.2008.916470
95. Richter R, Heege T, Kiselev V, Schläpfer D (2014) Correction of ozone influence on TOA radiance. Int J Remote Sens 35(8044):–8056. https://doi.org/10.1080/01431161.2014.978041
96. Shah M, Raval MS, Divakaran S, Patel P (2023) Study and impact analysis of data shift in deep learning based atmospheric correction. In IGARSS 2023 – 2023 IEEE International Geoscience and Remote Sensing Symposium, pp 6728–6731